

PERFORMANCE ANALYSIS OF HIGHLY EFFICIENT COMPACT DECODERS FOR SRAM ARRAYS

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**BACHELOR OF TECHNOLOGY
in
ELECTRONICS AND COMMUNICATION ENGINEERING**

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**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
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CERTIFICATE

This is to certify that the project report entitled ”***PERFORMANCE ANALYSIS OF HIGHLY EFFICIENT COMPACT DECODERS FOR SRAM ARRAYS***” being submitted by **S. DURGA BHAVANI, Y. VARSHITHA** and **Y. POOJITHA** in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in Electronics and Communication Engineering at V R Siddhartha Engineering College, affiliated to JNTUK, Kakinada, is a record of bonafide work carried out during the Academic Year 2025-26.

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DECLARATION

We hereby declare that the work presented in this major project titled “***PERFORMANCE ANALYSIS OF HIGHLY EFFICIENT COMPACT DECODERS FOR SRAM ARRAYS***” submitted towards the partial fulfillment of requirements for the award of the degree of **Bachelor of Technology** in Electronics and Communication Engineering in V R Siddhartha Engineering College, is an authentic record of our work carried out under the supervision of **Dr. Pratikhya Raut, Assistant Professor** in Electronics and Communication Engineering department, V R Siddhartha Engineering College Vijayawada.

The matter embodied in this report has not been submitted by us for the award of any other degree.

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ABSTRACT

This project presents the performance analysis of highly efficient compact decoders for SRAM arrays using the Gate Diffusion Input (GDI) technique. Conventional SRAM decoders suffer from high power consumption, increased delay, and larger area, which limit their scalability in modern VLSI systems. To address these challenges, a compact decoder based on GDI logic is designed and implemented. A 6T SRAM cell is modelled and simulated using Cadence Virtuoso, and SRAM arrays of sizes 2×2 and 4×4 are developed to evaluate system-level performance. The proposed GDI-based decoder significantly reduces transistor count, power consumption, and propagation delay compared to traditional CMOS designs. Performance analysis is carried out across multiple technology nodes (180 nm, 90 nm, and 45 nm) to verify scalability. Simulation results demonstrate that the proposed design achieves considerable improvements in power efficiency, speed, and area utilization. Hence, the GDI-based compact decoder is a promising solution for low-power, high-performance SRAM architectures in advanced VLSI applications.

Keywords:

2x2 SRAM Array, 4x4 SRAM Array, 45/90/180 nm Technology, 6T SRAM Cell, Cadence Virtuoso, Compact Decoder, GDI Technique, Low Power VLSI, Transistor Count Reduction

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LIST OF ABBREVIATIONS

SNM	Static Noise Margin
SRAM	Static Random Access Memory
CMOS	Complementary Metal–Oxide–Semiconductor
GDI	Gate Diffusion Input
VLSI	Very Large Scale Integration
VTC	Voltage Transfer Characteristics
nm	Nanometres
MGDI	Modified Gate Diffusion Input
TTL	Transistor–Transistor Logic
RTL	Resistor–Transistor Logic
DTL	Diode–Transistor Logic
ECL	Emitter–Coupled Logic

NOMENCLATURE

V_{DD}	Drain supply voltage
V_Q	Voltage at node Q
V_{QB}	Voltage at the complement node $\overline{Q}$
BL	Bit Line
$\overline{BL}$	Bit Line-bar
WL	Word Line
$'$	Complement of input
$.$	AND Operation
$+$	OR Operation

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

In modern digital systems, memory plays a crucial role in determining overall system performance, power consumption, and area efficiency. Among various memory types, Static Random Access Memory (SRAM) is widely used due to its high speed, low latency, and reliable operation. SRAM is extensively utilized in cache memories, embedded systems, and high-performance computing applications. An SRAM array consists of multiple memory cells along with peripheral circuits such as decoders, sense amplifiers, and write drivers. Among these, the decoder is a key component responsible for selecting the appropriate memory cell based on the input address. However, conventional CMOS-based decoders suffer from high power consumption, larger area, and increased delay, especially as technology scales down. To overcome these limitations, this project focuses on the design and performance analysis of compact decoders using the Gate Diffusion Input (GDI) technique. The GDI technique reduces transistor count, minimizes power consumption, and improves speed compared to traditional CMOS logic. In this work, a standard 6T SRAM cell is designed and simulated using Cadence Virtuoso. Further, 2×2 and 4×4 SRAM arrays are implemented, and their performance is analysed using both conventional and GDI-based compact decoders across different technology nodes (180 nm, 90 nm, and 45 nm). The proposed approach aims to achieve improved power efficiency, reduced delay, and optimized area, making it suitable for low-power VLSI applications.

1.2 OBJECTIVES

The main objectives of this project are:

- To design and validate a 6T SRAM cell using Cadence Virtuoso with DC and transient analysis.
- To design and simulate the SRAM array using Cadence Virtuoso environment.
- To implement hierarchical SRAM arrays (2×2 and 4×4) and evaluate their performance metrics.
- To develop a GDI-based compact decoder for reducing transistor count and power consumption.
- To compare the performance of conventional CMOS decoder with the proposed GDI-based decoder.
- To analyze the effect of technology scaling (180nm, 90nm, 45nm) on SRAM

performance.

- To ensure reliable operation of SRAM under different process variations.

1.3 MOTIVATION

With continuous CMOS scaling, the demand for low-power and high-speed memory systems has increased significantly. SRAM is widely used in cache and embedded applications due to its fast access time and reliable operation. However, conventional SRAM architectures face several challenges, including increased power consumption, delay, and area overhead, mainly due to inefficient decoder designs. The decoder plays a critical role in memory access, and traditional CMOS-based decoders suffer from high transistor count and switching activity, which degrade overall performance. Therefore, there is a strong need for optimized decoder architectures that can enhance SRAM efficiency. The Gate Diffusion Input (GDI) technique provides an effective solution by reducing transistor count, minimizing power consumption, and improving speed. Additionally, the requirement for scalable designs across different technology nodes further motivates the development of compact and efficient SRAM architectures. Hence, this work focuses on implementing a GDI-based compact decoder to achieve low-power, high-performance SRAM systems.

1.4 SCOPE OF THE PROJECT

This project focuses on the design and performance analysis of SRAM arrays integrated with compact decoders using the Gate Diffusion Input (GDI) technique. The scope includes the modelling and simulation of a standard 6T SRAM cell, followed by the development of small-scale SRAM arrays such as 2×2 and 4×4 configurations. Both conventional CMOS-based decoders and GDI-based compact decoders are designed and implemented to evaluate their impact on overall memory performance. The analysis is carried out using Cadence Virtuoso, considering key performance parameters such as power consumption, propagation delay, and transistor count. The study also includes evaluation across different technology nodes (180 nm, 90 nm, and 45 nm) to analyse scalability and efficiency. The primary focus of this project is to achieve low-power, high-speed, and area-efficient SRAM designs through decoder optimization, making it suitable for modern VLSI applications.

1.5 THESIS ORGANIZATION

This thesis is structured into six chapters, each with a specific focus. A brief summary of these chapters is presented below.

Chapter 1: serves as an introduction to the thesis, highlighting the main objectives,

motivation, research problems, and key contributions of the work.

Chapter 2: provides an overview of fundamental concepts and background knowledge related to SRAM design, CMOS technology, and low-power VLSI techniques. It also reviews existing literature on SRAM architectures and decoder designs, along with their limitations.

Chapter 3: presents the design and implementation of the SRAM array architecture. It includes the structure of the 6T SRAM cell, organization of 2×2 and 4×4 SRAM arrays, and the design of peripheral circuits such as decoders, sense amplifiers, and write drivers.

Chapter 4: describes the proposed approach adopted in this work. It focuses on the design and implementation of a compact decoder using the GDI technique, along with methodologies for reducing power consumption, minimizing transistor count, and improving overall performance.

Chapter 5: discusses the simulation results and analysis. It provides a detailed evaluation of parameters such as power consumption, delay, and area, along with a comparative study between conventional CMOS designs and the proposed GDI-based approach.

Chapter 6: concludes the thesis and outlines future research directions, including scalability to larger SRAM arrays, further power optimization techniques, and advancements in low-power VLSI design.

CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION

The literature review shows that extensive research has been carried out on SRAM cell design, performance optimization, and technology scaling. Most studies focus on improving parameters such as SNM, delay, and power consumption at the cell level, along with exploring advanced SRAM structures like 10T and 12T cells for better stability and leakage control. Additionally, SRAM array implementations highlight the growing impact of peripheral circuits on overall performance. However, it is observed that conventional CMOS-based decoders contribute significantly to power consumption and area overhead, especially in larger arrays. Although some work has been done on compact decoder designs, limited research exists on integrating low-power techniques like the GDI method with SRAM arrays.

2.2 LITERATURE REVIEW

Y. Syamala *et al.* [1] proposed low-power CMOS logic circuits using the Gate Diffusion Input (GDI) technique. Their approach significantly reduces transistor count, switching activity, and dynamic power consumption compared to conventional CMOS logic. This technique is highly suitable for compact decoder design in SRAM arrays due to its area and power efficiency.

G. Shivaprakash and D. S. Suresh [2] designed a low-power 6T SRAM cell optimized for high-speed applications. Their work focuses on reducing access delay and improving stability, indicating that efficient peripheral circuits such as decoders are essential for achieving faster memory access.

S. Rath *et al.* [3] analyzed the performance of 6T SRAM cells across different technology nodes. The study highlights variations in power, delay, and noise margins, emphasizing the need for scalable and efficient decoder circuits compatible with advanced technologies.

R. Ruchi and S. Dasgupta [4] developed a compact analytical model to evaluate write static noise margin (WSNM) in SRAM cells. Their work ensures reliable write operations, which depend on accurate word-line selection by decoder circuits.

B. Kaleeswari *et al.* [5] designed and implemented an 8T SRAM cell in memory arrays to enhance read stability and reduce leakage power. Their findings indicate that as memory arrays grow, decoder efficiency becomes critical for maintaining overall performance.

J. Shrivastava *et al.* [6] investigated the impact of design parameters such as supply voltage and transistor sizing on SRAM bit cell performance. The study shows that delay and power consumption are influenced not only by the cell but also by supporting circuits like decoders.

S. Singh *et al.* [7] proposed a low-power 1 KB SRAM array using a compact 7T SRAM cell. Their design reduces leakage power and improves efficiency, suggesting that optimized decoder architectures can further enhance memory performance.

A. Agal *et al.* [8] presented the design and analysis of a 6T SRAM cell with emphasis on minimizing power and area. Their work highlights the importance of integrating efficient peripheral circuits, including decoders, to achieve overall system optimization.

S. Joshi *et al.* [9] implemented a low-power CMOS SRAM cell using 90 nm technology in Cadence tools. The study demonstrates how technology scaling affects power consumption and performance, requiring efficient decoder designs.

P. Agarwal *et al.* [10] proposed a 7T SRAM cell using various circuit techniques to reduce dynamic and leakage power. Their approach improves write ability and stability, indirectly emphasizing the need for efficient address decoding.

S. Saun *et al.* [11] analyzed 6T SRAM cells across different CMOS technologies, focusing on stability metrics such as static noise margin. Their results highlight that peripheral circuits like decoders must also scale efficiently.

U. Vasudevan *et al.* [12] implemented digital circuits using both GDI and CMOS techniques. Their comparison shows that GDI-based designs offer reduced power and area, making them suitable for compact decoder implementation.

The work in [13] proposed a 10T SRAM cell with improved read performance and expanded write margin. The design enhances reliability, which depends on precise control signals generated by decoder circuits.

S. Mutum *et al.* [14] performed a comparative analysis of 6T SRAM cells in 180 nm and 90 nm technologies. Their study highlights scaling challenges, reinforcing the need for optimized decoder designs in advanced nodes.

B. V. Garidepalli *et al.* [15] designed a low-power 4 kb SRAM array using GNR-FET and CMOS technologies. Their work demonstrates improved energy efficiency and reduced delay, emphasizing the importance of efficient decoding in large arrays.

Xue *et al.* [16] analyzed a 32×32 SRAM array for low-power applications, focusing on optimizing read/write operations. Their work suggests that decoder performance significantly impacts array efficiency.

D. Mitta [17] studied SRAM cell design across different technology nodes, analyzing

trade-offs between power, delay, and stability. This highlights the need for compatible and efficient decoder circuits.

N. Dasharath *et al.* [18] proposed efficient 2×2 and 4×4 SRAM arrays using compact decoders. Their design achieves reduced power consumption and area, making it highly relevant to compact decoder performance analysis.

Namgiri [19] designed an 8×8 SRAM memory array using 45 nm technology operating at 100 MHz. The study demonstrates improved speed and performance, relying on relevant to compact decoder performance analysis.

Sabitabratta Bhattacharya *et al.* [20] compared SRAM performance across 45 nm, 90 nm, and 180 nm technologies. Their work highlights variations in delay and power, emphasizing the need for optimized decoder circuits.

The work in [21] proposed SRAM cells with improved read and write performance using assist techniques. These techniques enhance stability, which depends on proper decoder-driven control signals.

The study in [22] introduced a 12T SRAM cell with leakage power control techniques to improve energy efficiency. This further emphasizes the importance of low-power peripheral circuits such as compact decoders.

From the above literature, it is evident that most existing research focuses on SRAM cell design, stability, and power optimization. However, limited work has been dedicated specifically to the performance analysis of compact decoders. Therefore, this work aims to analyze and optimize compact decoder architectures to improve SRAM array performance in terms of power, delay, and area.

2.3 PROBLEM STATEMENT

The advancement of CMOS technology has enabled the development of high-density SRAM arrays; however, it has also introduced critical challenges related to power consumption, delay, and stability. As technology scales down, SRAM cells experience reduced noise margins and increased leakage currents, which affect reliable operation. Moreover, the performance of SRAM arrays is not only dependent on the memory cell but also significantly influenced by peripheral circuits such as decoders. Conventional decoder designs based on CMOS logic involve a high number of transistors, resulting in increased switching power, larger silicon area, and higher propagation delay. These limitations become more prominent in larger memory arrays, making traditional designs unsuitable for modern low-power VLSI applications. Additionally, existing research primarily focuses on SRAM cell improvements, while decoder optimization has not

been extensively explored using alternative low-power design techniques in VLSI digital Systems.

2.4 RESEARCH GAP IDENTIFIED

Most existing research focuses on improving SRAM cell performance in terms of stability, power, and delay using various design techniques and technology scaling. However, limited attention has been given to peripheral circuits, especially decoders, which significantly influence overall SRAM array performance. Conventional CMOS decoders suffer from high transistor count, increased power consumption, and larger area. Although some compact decoder designs exist, the use of low-power techniques like Gate Diffusion Input (GDI) in decoder design is still limited. Additionally, there is a lack of integrated studies combining compact decoders with SRAM arrays across different technology nodes. Hence, there is a need to design a low-power, area- efficient, and high-speed GDI-based decoder and analyze its impact on SRAM array performance.

CHAPTER 3

DESIGN ARCHITECTURE

3.1 APPROACH

The proposed work aims to design a power-efficient and area-optimized SRAM architecture by using compact decoders based on the GDI technique. Unlike conventional CMOS decoders, GDI-based designs require fewer transistors, which reduces power consumption, delay, and chip area. The proposed decoder is integrated with 6T SRAM cells in 2×2 and 4×4 arrays, along with necessary peripheral circuits. A comparative analysis with traditional decoders across different technology nodes shows that the GDI-based approach significantly improves overall performance and scalability of SRAM systems.

3.2 METHODOLOGY

Figure 3.1 Illustrates the methodology of this project begins with the design of a 6T SRAM cell using CMOS technology. The performance parameters such as Static Noise Margin (SNM) and delay are analyzed under different technology nodes. Supporting circuits including decoders, sense amplifiers, and write drivers are designed and integrated to form a complete SRAM array. Initially, SRAM arrays are implemented using conventional CMOS decoders and their read/write delays are analyzed. Further, compact decoders are designed using the Gate Diffusion Input (GDI) technique to reduce transistor count and power consumption. These decoders are integrated into the SRAM array and performance is evaluated. The proposed design is analyzed in terms of power, delay, and area, and compared with conventional designs. Both pre-layout and post-layout simulations are carried out to validate the performance. Finally, the results are analyzed to demonstrate the effectiveness of the proposed approach.

3.2.1 Schematic Design of 6T SRAM Cell

The methodology begins with the schematic design of the 6T SRAM cell, constructed using two cross-coupled CMOS inverters and a pair of access transistors controlled by the word line. The design is created in Cadence Virtuoso, where the connectivity of bit lines and control signals is validated to guarantee correct operational behavior before proceeding to further analysis. As shown in Figure 3.2 illustrate the detailed transistor-level and logical representations of the 6T SRAM cell The transistor schematic consists of two cross-coupled CMOS inverters that form the bistable storage latch.

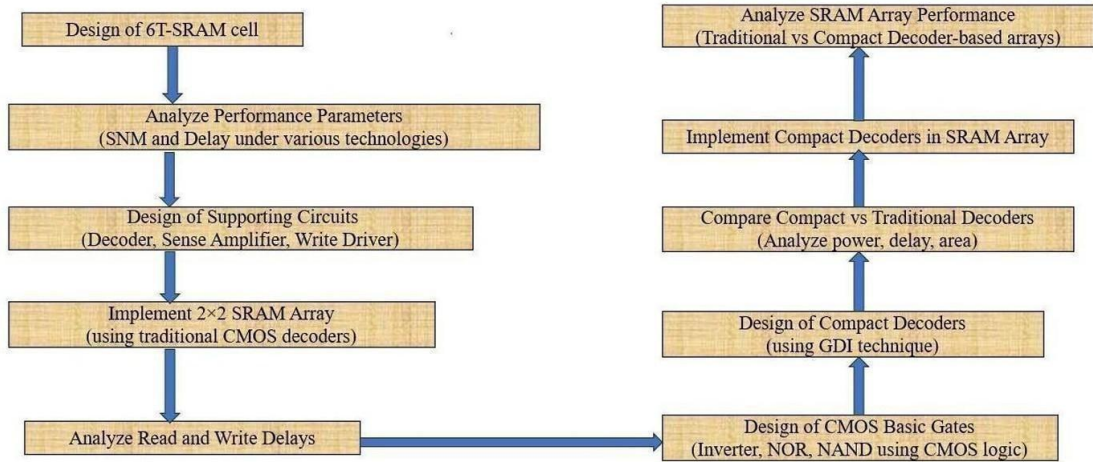


Fig. 3.1 Flow Chart of Methodology

The logical diagram provides an abstract view of the same structure using two inverters and access switches, highlighting the functional behaviour of data storage and access without focusing on device-level complexity. Figure 3.3 shows the schematic implementation of the 6T SRAM cell designed in the Cadence Virtuoso environment. The schematic includes the two cross-coupled CMOS inverters forming the storage latch and the two access transistors that connect the internal nodes to the bit lines during read and write operations. An SRAM (Static Random Access Memory) cell is a basic memory unit used to store 1 bit of data. It is called static because it retains its value as long as power is supplied, without needing refresh. Unlike dynamic memory, SRAM does not require periodic refreshing, as it maintains its stored value using a feedback mechanism as long as the power supply is present. In the write operation, the access transistors must be strong enough to overwrite the previous data stored in the cell. This requires proper transistor sizing to ensure successful write operation without increasing power consumption significantly. The write margin is another important parameter that defines the ease of writing new data into the cell. Leakage power is a major concern in modern SRAM designs, especially as technology scales down to deep submicron levels. Subthreshold leakage and gate oxide leakage contribute to power dissipation even when the cell is not actively switching. Various techniques such as transistor sizing, power gating, and use of alternative logic styles are employed to reduce leakage power. The performance of the SRAM cell is also influenced by process variations, temperature changes, and supply voltage fluctuations. These factors can affect the reliability and stability of the cell, making it essential to design robust SRAM architectures.

BL' which is detected by the sense amplifier.

Write Operation: During the write operation, the word line (WL) is activated to enable the access transistors. The input data is applied to the bit lines (BL and BL'). Depending on the input value (0 or 1), one of the inverters is forced to change its state. The cross-coupled inverter pair then reinforces the new value and stabilizes the stored data.

3.2.3 DC Analysis of 6T SRAM Cell

The DC analysis of the 6T SRAM cell is performed to evaluate its Static Noise Margin (SNM), which is a key indicator of cell stability. The butterfly curve is obtained by plotting the voltage transfer characteristic (VTC) of one inverter against the mirrored VTC of the second inverter. The two lobes formed in this plot represent the bistable operating points of the cell. From the Figure 3.4 the largest square that can fit between the two lobes determines the SNM value, indicating the maximum tolerable noise before the cell flips its stored state. This analysis is essential for understanding how supply voltage, device sizing, and technology scaling affect the stability and robustness of the SRAM cell.

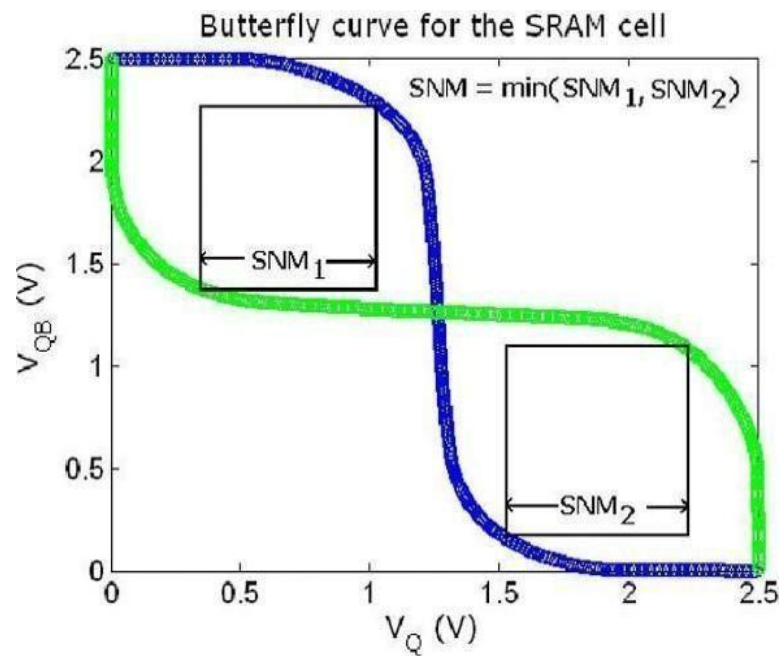


Fig. 3.4 V_Q vs V_{QB} for SRAM Cell

Figure 3.5 explains the DC analysis of the 6T SRAM cell using the butterfly curve. The graph represents the voltage transfer characteristics of the two cross coupled inverters in the SRAM cell. The intersection of these curves forms the butterfly shape, which is used to determine the Static Noise Margin (SNM). The maximum square that can be embedded between the two curves indicates the SNM value, defines the stability of cell.

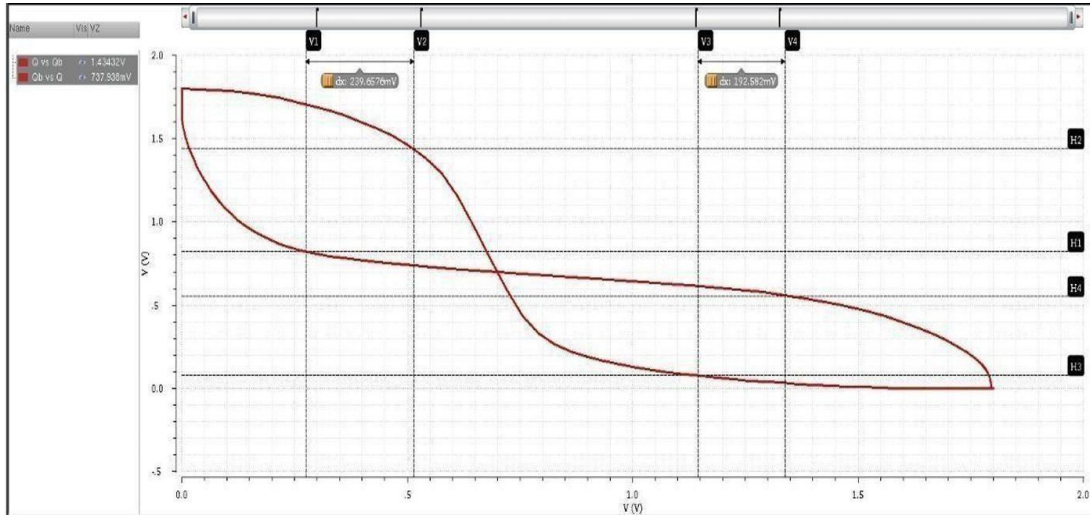


Fig. 3.5 V_{dd} vs SNM in SRAM Cell in Hold Operation

3.2.4 Implementation of SRAM Array

SRAM (Static Random Access Memory) is a volatile memory that stores data using bistable latching circuits. It operates faster and consumes less power than DRAM since it does not require refresh operations. Due to its high speed and low latency, SRAM is mainly used in cache memory, registers, and buffer storage. An SRAM cell typically consists of six transistors arranged as cross-coupled inverters along with access transistors. SRAM arrays are organized in rows and columns, where each cell stores one bit of data. Key components include word lines (row selection), bit lines (data transfer), row/column decoders, sense amplifiers, and write drivers, all contributing to overall performance. The performance of SRAM mainly depends on power consumption and delay. The total delay includes word line activation time, bit line discharge time, sense amplifier response, and write operation time. Efficient design requires optimizing both delay and power simultaneously. This study analyzes 2×2 (4-bit) and 4×4 (16-bit) SRAM arrays. The 2×2 array is simple with low power consumption, while the 4×4 array is more complex due to increased bit line capacitance, requiring better design techniques to maintain stability and performance.

Required Peripheral Circuits for Implementation

- **Decoder:**

- Selects the word line based on the input address.
- Enables access to the targeted cell row during read/write operations.

- **Sense Amplifier:**

- Detects and amplifies the small voltage difference on the bit lines.
- Ensures fast and reliable data sensing.

- **Write Driver:**

- sufficient drive strength to overwrite the existing data.
- Forces correct voltage levels on bit lines.

During operation, when the word line is activated, the selected SRAM cell interacts with the bit lines through access transistors. In read mode, data is sensed by the sense amplifier, while in write mode, new data is written into the cell through the write driver. When the word line is deactivated, the cell enters the hold state, maintaining its stored value using internal feedback. Overall, the performance of the SRAM system depends on the efficient coordination of these components, with key parameters such as power consumption, delay, and area being critical for optimization. In this project, the use of compact decoders based on the GDI technique helps in reducing transistor count and improving overall efficiency.

3.2.5 Decoder

The decoder system executes activation of a single word line together with its individual bit line according to an input address. The decoder chooses particular output lines to receive data entry commands. Three memory sections in addition to the decoder work together as an addressing system to manage computer core memory data storage. The proper operation of the decoder circuit requires implementing sixteen transistors. The basic operation of the decoder appears in Figure 3.6 through a visual representation of its executing process. Using a 1:2 configuration decoder enables operation throughout small-scale memory systems due to its 2×2 array structure. A 2:4 decoder serves as an essential part in memory addressing for 4×4 SRAM arrays to achieve efficient functionality. The 1:2 decoder operates through two output lines controlled by a single input by using the input address for selecting appropriate outputs according to Figure 3.7. The comprehensive operation of an SRAM array relies on proper functioning between its row and column decoders. The row decoder energizes different word lines through its operations which allows the system to access the intended memory cell for read and write operations. A reliable memory operation system results from proper decoding structure implementation in SRAM. It converts binary input signals into a unique output line, enabling selection of a specific word line in the SRAM array.

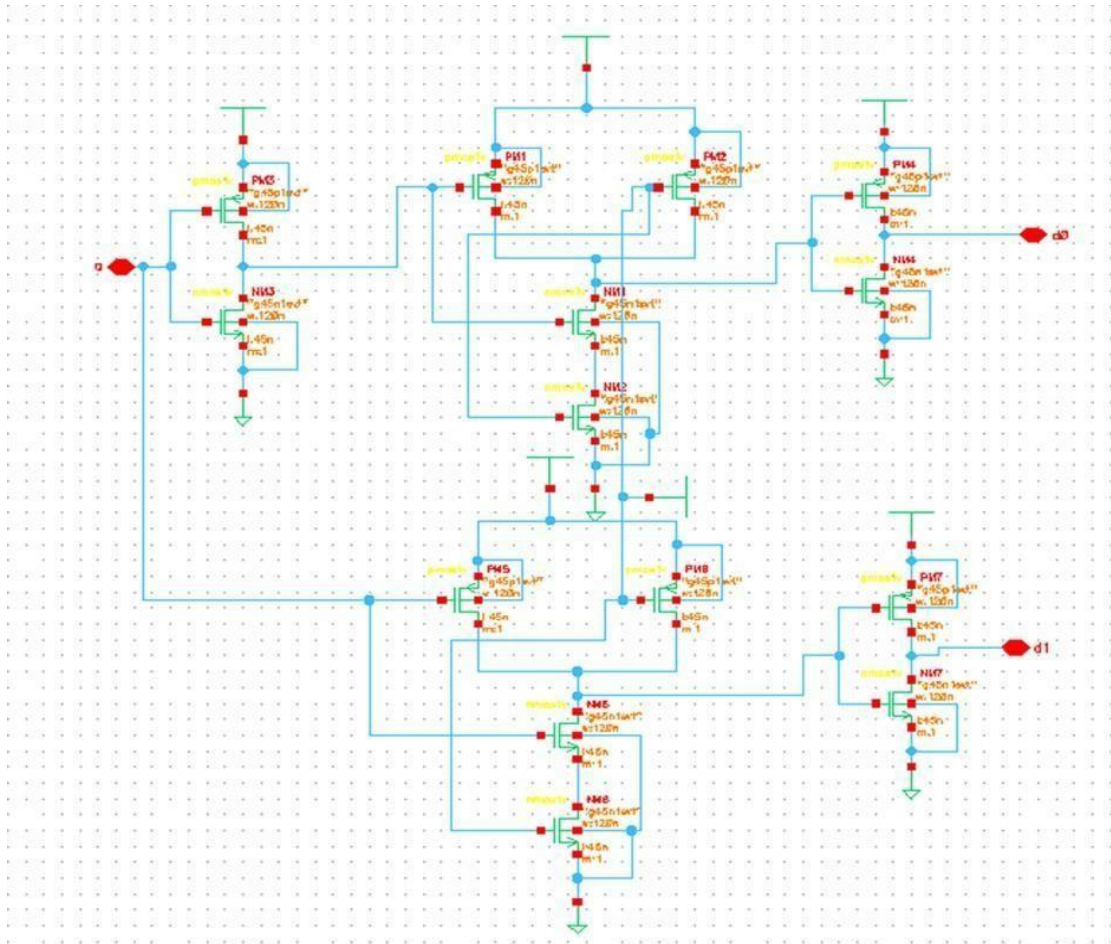


Fig. 3.6 Schematic Diagram of Decoder



Fig. 3.7 Block Diagram of Decoder

Transient Response of Decoder:

Figure 3.8 explains the transient response of the decoder, which shows how the decoder output changes with time for different input signals. It illustrates the switching behavior, propagation delay, and signal transitions during operation. This analysis helps in evaluating the speed and performance of the decoder in terms of delay and stability. transient analysis of the decoder is essential for evaluating its delay, switching characteristics, and overall impact on SRAM performance. The power consumption during switching can be analyzed, as the decoder consumes dynamic power when transitioning between logic states.

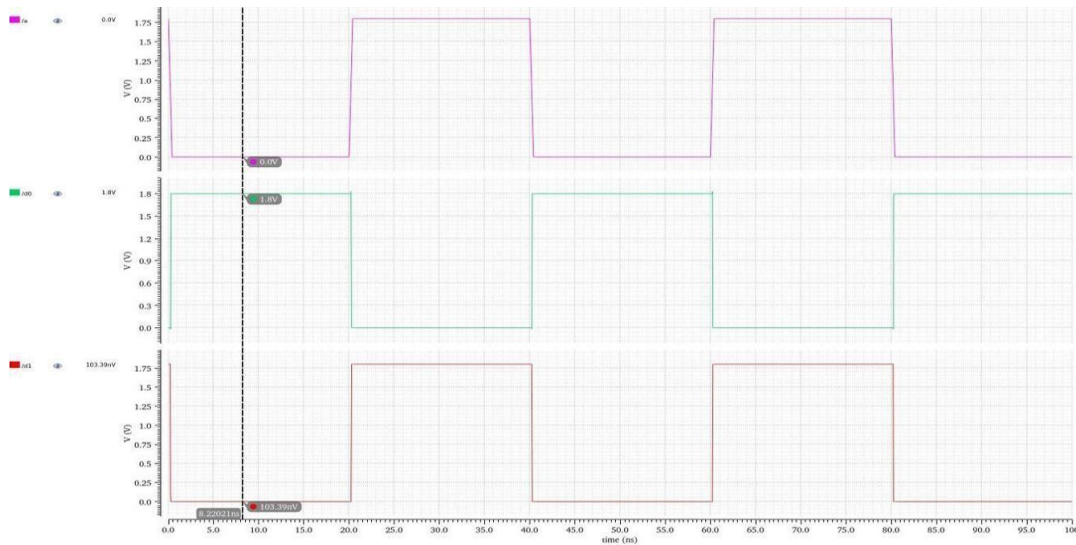


Fig. 3.8 Transient Response of Decoder

3.2.6 Sense Amplifier

The sense amplifier performs read operations by detecting the stored value from an SRAM cell while moving it to the output stage. A sense amplifier integrated within the read circuitry detects tiny bit line voltage changes through amplification and transforms them into logical signal outputs for accurate interpretation. The data retrieval process requires amplification to achieve accurate results for read operations. The sense amplifier block diagram shown in Figure 3.10 contains three inputs and one output pin. During a read cycle, very small voltage differences appear on the bit lines due to the stored data. These differences are often too weak to be directly interpreted by digital circuits. The system allows BL (Bit Line) and BLB (Bit Line Bar) pins access the data while SE (Sense Enable) is switched on. The amplified output of the amplifier appears at Vout providing readable interpreted data values to the external circuit. The read operation functions effectively because the sense amplifier maintains signal clarity through its ability to neutralize signal noise and disturbances found in bit lines. A detailed depiction of sense amplifier design including its connections can be found in Figure 3.9.

Transient Response of Sense Amplifier:

Transfer of data from other read circuitry components to the sense amplifier shows how accurate signal amplification occurs. The sense amplifier's processing activity with respect to bit line voltage fluctuations is displayed through Figure 3.11 along with a depiction of the output's transient behavior throughout time. The momentary output response shows how the amplifier functions in speed and data accuracy during retrieval operations. The sense amplifier detects this difference and quickly amplifies it to either logic '1' or logic '0'. This process ensures fast and accurate data retrieval.

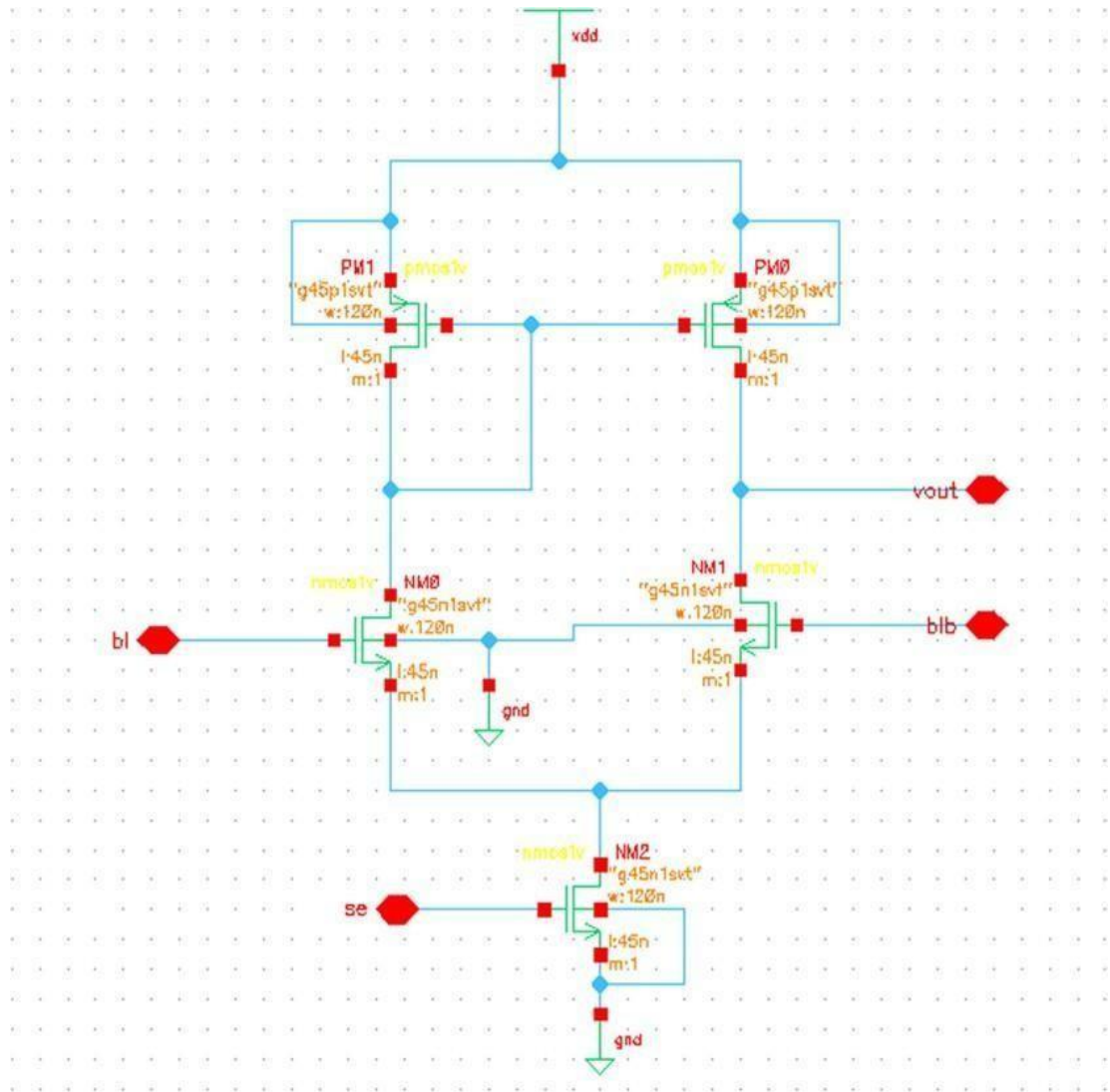


Fig. 3.9 Schematic Diagram of Sense Amplifier

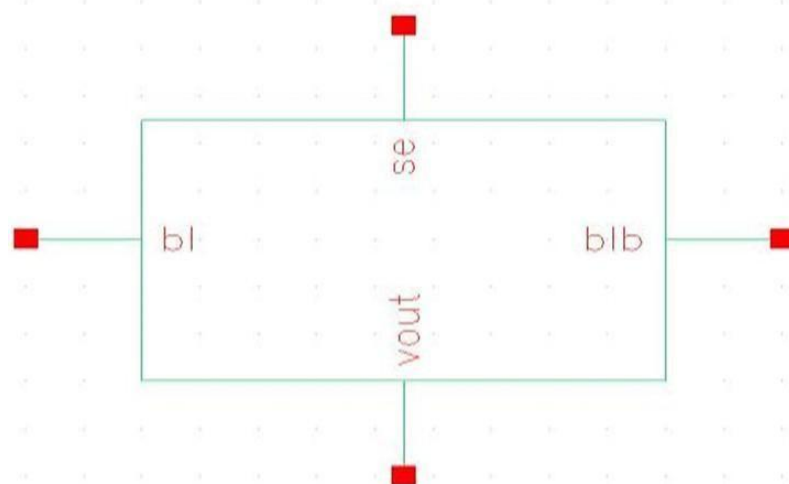


Fig. 3.10 Block Diagram of Sense Amplifier

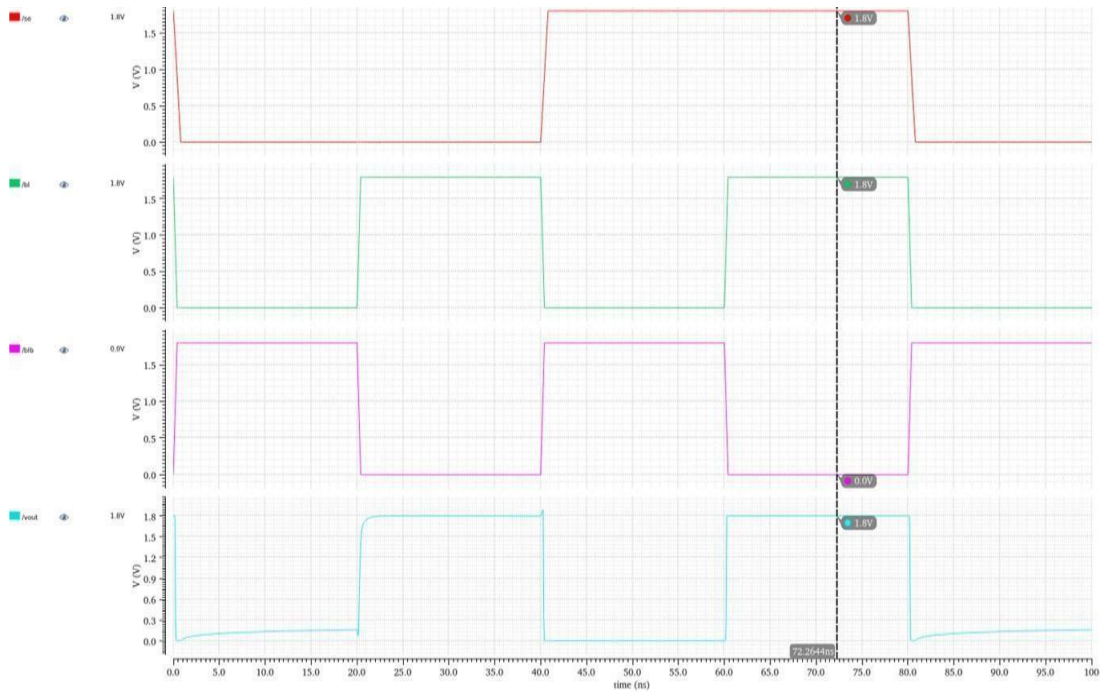


Fig. 3.11 Transient response of Sense Amplifier

3.2.7 Write Driver

The write driver performs an essential function in storing data on the bit lines BL and BL' through the writing process. The write driver controls the bit line voltages to guarantee correct storage of data in the SRAM cell. The write drivers act as core elements for data storage since they determine data movement and preservation patterns in the memory system. Through the write driver's operation, it applies desired data to one bit line without changing the values stored on the other bit line. Correct information storage in SRAM cells becomes possible through controlled modifications of the SRAM circuit. Figure 3.12 and Figure 3.13 are the schematic and block diagrams of sense amplifier. A well-controlled write driver operation stands vital for both data protection and memory system performance stability. The performance of the write driver in SRAM architecture determines how reliable the memory system operates. A properly operating write driver provides complete success in every successful write operation with zero errors. The write driver's main purpose in SRAM cells is to establish the correct logic values on the cells during write operations because these cells employ bistable latches for data retention. A well-paired operation between one bit line data forcing and simultaneous quieting of the other results in state transformation of the cell. 17 The driver operates in a previously engineered format to execute writing operations swiftly while maintaining power efficiency. The system must drive sufficient voltages which guarantee data overwrite and reliable operation of existing information.

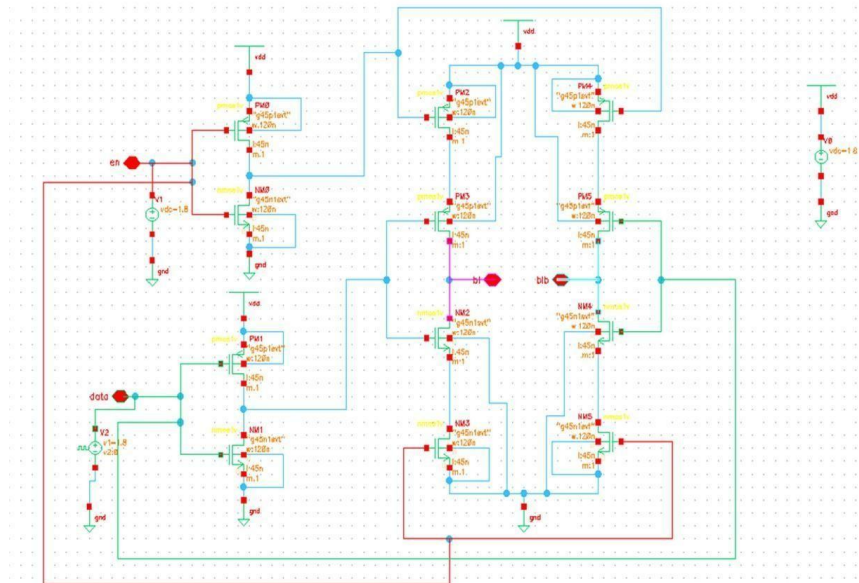


Fig. 3.12 Schematic Diagram of Write Driver

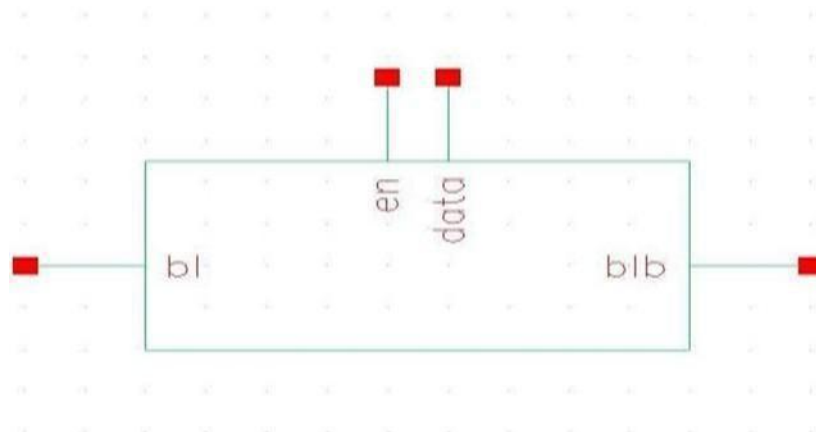


Fig. 3.13 Block Diagram of Write Driver

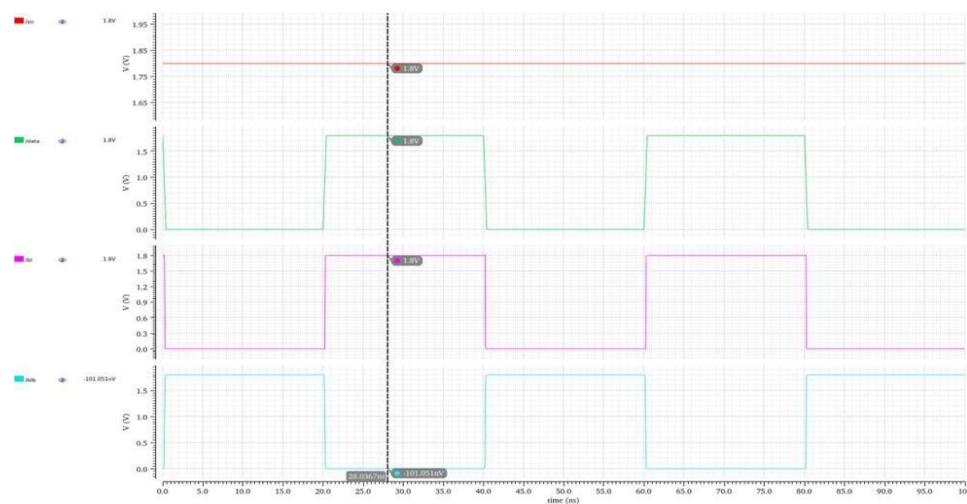


Fig. 3.14 Transient response of write driver

Transient Response of Write Driver:

Figure 3.14 explains the transient response of the write driver, which shows how the write driver signals vary with time during a write operation. It illustrates how data is driven onto the bit lines (BL and BL') to store a value in the SRAM Arrays.

3.2.8 2 x 2 SRAM Array

A 2x2 SRAM array is defined as a memory block consisting of four memory cells, supported by 1 : 2 row and 1 : 2 column decoders. Each 2 × 2 array has the capacity to store four bits, considering that each SRAM cell stores one bit of data. The selection of memory cells is performed using row and column decoders. Each cell consists of a bistable latch implemented using six transistors (6T), located at the intersection of a row and a column. During the write enable condition, the write driver performs the data storage operation in the selected memory cell. The input address is applied to activate the corresponding word line (WL), which selects the desired row. The 2 × 2 SRAM array allows read and write operations through bit lines (BL and BL'), while the address lines determine the selected row and column. Due to its fast operation and ability to retain data as long as power is supplied, SRAM is widely used in small cache memory blocks and registers. The functionality of the 2 × 2 SRAM array is based on decoder-controlled selection of memory cells through row and column paths. Stable data retention is ensured by bistable latches, which hold the stored data until it is intentionally modified. The sense amplifier detects small voltage differences on the bit lines during read operations and amplifies them to produce accurate output data. This amplification ensures reliable data retrieval from the memory cells.

Write Operation

To store data in a specific SRAM cell, the row and column decoders receive the input address. These decoders identify the exact location of the target cell within the array. Once the cell is selected, the write driver transfers the input data to the corresponding bit lines and forces the SRAM cell to store the new value. The write driver plays a key role in ensuring correct and reliable data storage.

Read Operation

During the read operation, the input address is provided to both row and column decoders to select the desired cell. The selected cell develops a small voltage difference on the bit lines. The sense amplifier detects this difference and amplifies it to produce the output. The retrieved data is available at the output node (V_{out}). Accurate data retrieval is achieved through the combined operation of decoders and the sense amplifier.

As shown in Figure 3.15, the schematic diagram of a 2×2 SRAM array consists of four SRAM cells arranged in two rows and two columns. Each cell is connected through word lines (WL) for row selection and bit lines (BL and $\overline{BL}$) for data transfer. The array also includes row decoders, sense amplifiers, and write drivers to perform read/write operations efficiently.

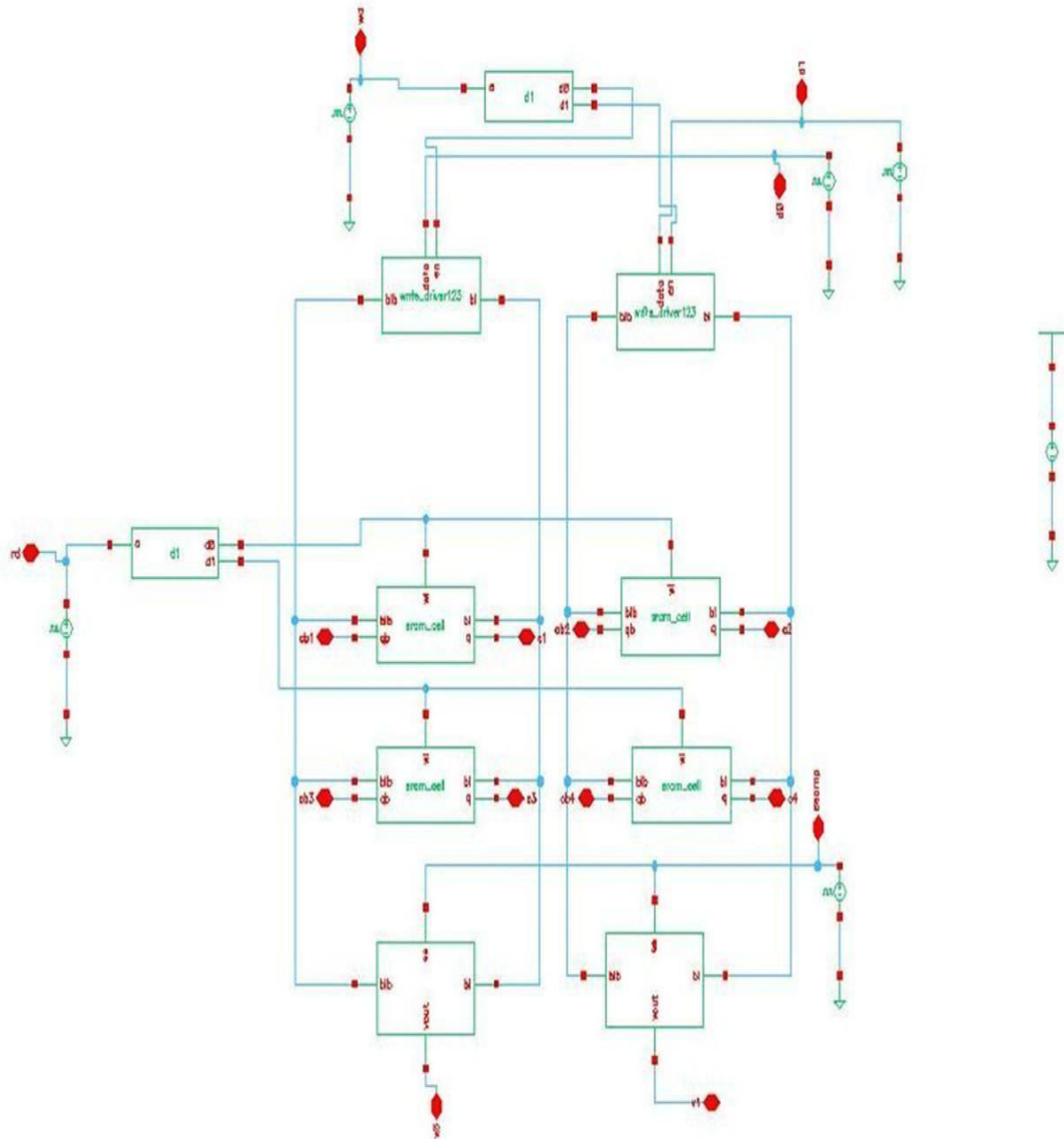


Fig. 3.15 Schematic Diagram of 2x2 SRAM Array

Transient response of 2x2 SRAM Array:

Figure 3.16 explains the transient response of the 2×2 SRAM array, which shows how the voltages of word lines and bit lines vary with time during read and write operations. It illustrates the dynamic behavior of all four cells, including data access and storage. This waveform helps evaluate overall array performance in terms of delay, power consumption, and stability.

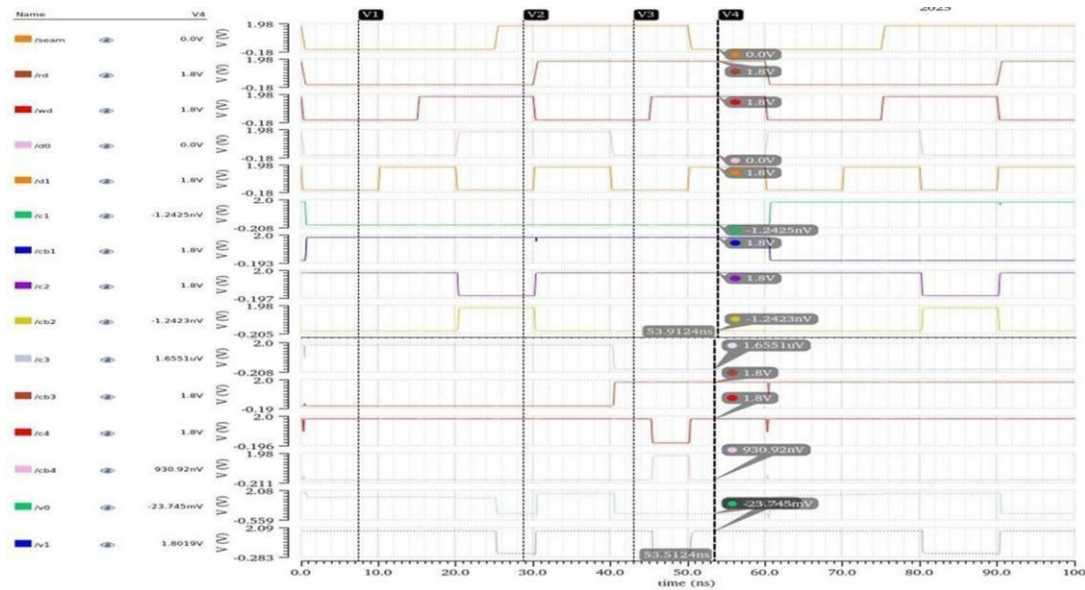


Fig. 3.16 Transient response 2x2 SRAM Array

Table 3.1 represents the transient analysis of a 2×2 SRAM array, showing the behavior of different cells during read and write operations. The inputs RD (Row Decoder) and WD (Write Decoder) select a particular cell (C1–C4). This table helps in understanding how different combinations of decoder inputs control cell selection and data transfer, validating correct SRAM functionality and timing behavior.

Table 3.1 Transient Analysis of 2×2 SRAM Array

RD	WD	Cell	d0	d1	Write	Read
0	0	C1	0	0	0 (d0)	0 (v0)
0	1	C2	1	0	0 (d1)	0 (v1)
1	0	C3	1	1	1 (d0)	1 (v0)
1	1	C4	0	1	1 (d1)	1 (v1)

3.2.9 4 x 4 SRAM Array

Figure 3.17 Illustrates the 4×4 SRAM array contains sixteen memory cells structured into a four-row four-column type format. The expanded 4×4 SRAM array demonstrates higher complexity since both power consumption and delay performance requirements have become more demanding. An elevated bit line capacitance influences both read and write operations thus demanding stable sense amplifiers. Analyzing power consumption ran through the Cadence tool during the

evaluation at 45nm, 90nm and 180nm technology nodes. The reduction in technology size leads to lower dynamic power consumption but leakage power becomes more dominant at smaller technology nodes. Power variations between these nodes must be comprehended to maximize SRAM performance for various applications.

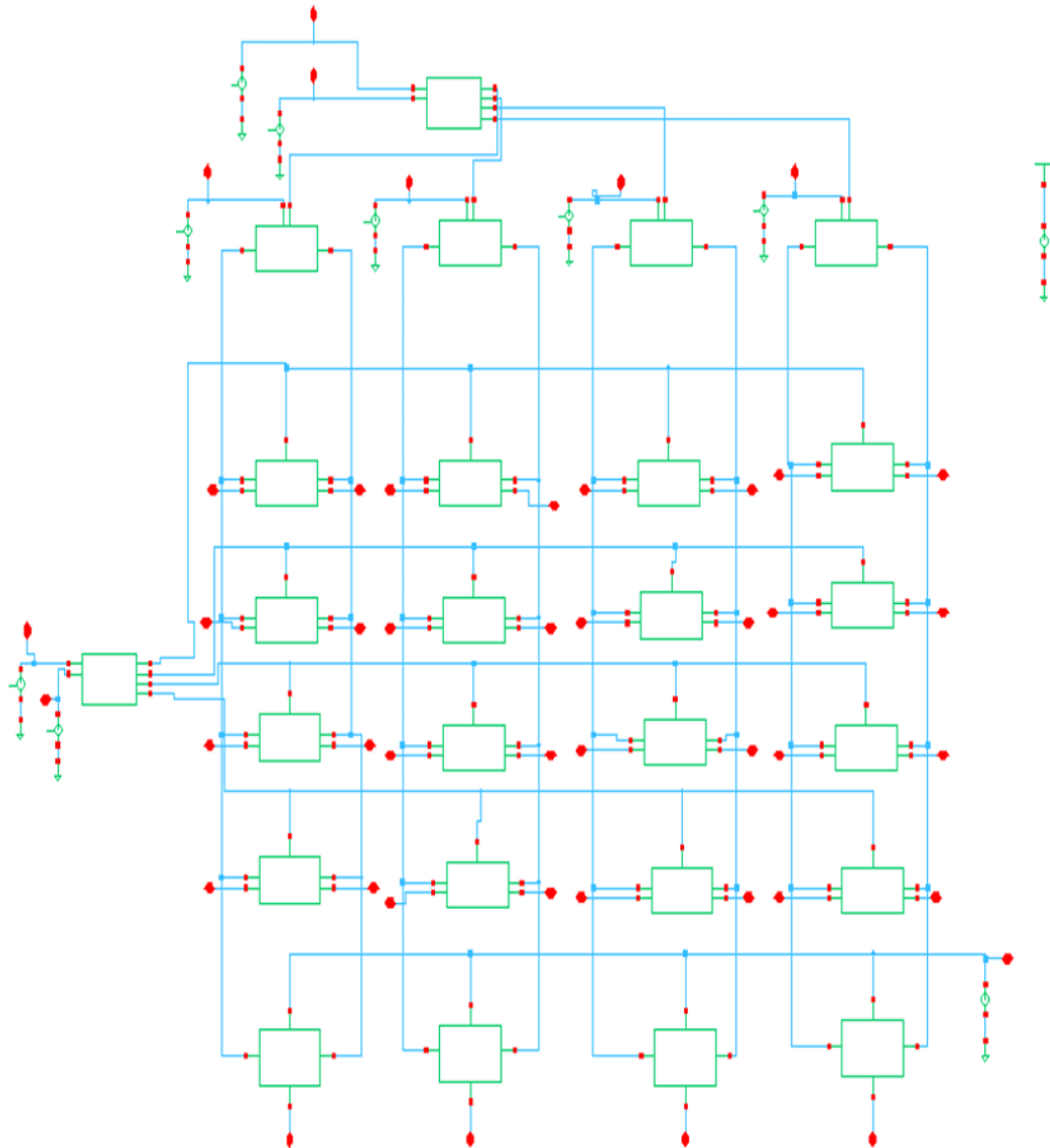


Fig. 3.17 Schematic Diagram of 4x4 SRAM Array

3.2.10 Power Efficiency

The power measurements for 2×2 and 4×4 SRAM arrays were conducted using the Cadence tool. The initial step required us to extract transient data from the array by accessing Cadence's appropriate viewing area. You accessed the browser type section inside the window to generate the power graph. The power measurement process ended through data transfer of the power graph to the calculator application. The methodical

procedure allowed us to properly examine SRAM array power usage and establish correct power measurement results. The analysis procedure revealed different power characteristics that emerged when examining arrays of various sizes and layouts. The Cadence power analysis tool delivered fundamental circuit behavioral information that deepened our knowledge about how SRAM arrays use power at different technology nodes. A smaller technology scale leads to decreased dynamic power usage because transistor capacitance values shrink. Broadened transistors tend to reduce their capacitance value thus decreasing their dynamic power consumption levels. Advanced technology nodes cause the supply voltage to decline as a result which leads to even lower power consumption. Power efficiency optimization occurs in SRAM arrays because these reductions become possible with technological advances.

3.2.11 Power Analysis of SRAM Arrays

The Table 3.2 shows the power consumption of 2×2 and 4×4 SRAM arrays for different technology nodes. As the technology scales from 180 nm to 45 nm, power consumption decreases significantly due to reduced transistor size and improved efficiency.

Table 3.2 Power Comparison of 2×2 and 4×4 CMOS SRAM Arrays

Technology	Power in 2×2 SRAM Array (μW)	Power in 4×4 SRAM Array (μW)
45 nm	197	413
90 nm	200	589
180 nm	450	932

Also, the 4×4 SRAM array consumes more power than the 2×2 array because it has more cells and higher capacitance. This comparison highlights the importance of technology scaling and array size in optimizing SRAM power performance.

3.3 SUMMARY

Chapter 3 presents the design architecture of the SRAM system, focusing on the implementation of a 6T SRAM cell and its integration into 2×2 and 4×4 arrays. The operation of the SRAM cell in hold, read, and write modes is explained, along with DC analysis using the butterfly curve to evaluate stability through SNM. Key peripheral circuits such as the decoder, sense amplifier, and write driver are designed to enable efficient memory operations. Transient analysis is performed to study timing and switching behavior. The chapter also analyzes power consumption across different technology nodes, showing improved efficiency with scaling.

CHAPTER 4

PROPOSED APPROACH

4.1 INTRODUCTION

The characteristics of a logic gate depend on its construction technology which determines the voltage requirements together with fan-out speed and power usage. The group of logic families consists of RTL, DTL, TTL, MOS and ECL. Several gates have become obsolete and designers no longer use them in their designs. The TTL along with the ECL and MOS and CMOS logic families represent well-liked technological choices that appear commonly within digital circuits. Complementary Metal Oxide Semiconductor exists under the name CMOS. The construction of CMOS logic gates requires simultaneous use of NMOS with PMOS transistors. Implementing MOS transistors in logic gate operations results in the production of a switching system. CMOS logic incorporates both NMOS and PMOS transistors for its design. From the Figure 4.1 the pull-up function depends on the PMOS transistors while the pull-down operation relies on NMOS transistors. CMOS logic gates and circuit achieve low static power usage since their design combines both NMOS and PMOS transistors.

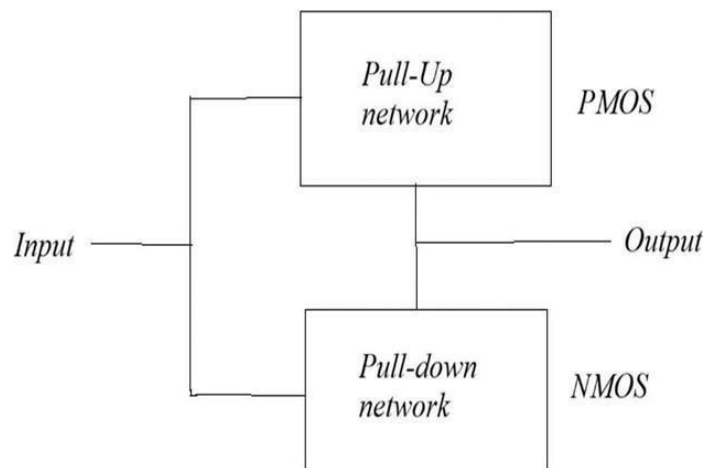


Fig. 4.1 Pull-up and pull-down network

CMOS (Complementary Metal-Oxide-Semiconductor) logic is widely used in digital circuit design due to its low power consumption and high reliability. It consists of two complementary transistor networks: the pull-up network (PMOS) and the pull-down network (NMOS), it achieves efficient switching with minimal power consumption, making it ideal for modern VLSI systems. CMOS uses complementary PMOS and NMOS networks to produce a reliable output with low power consumption.

4.2 IMPLEMENTATION OF LOGIC GATES USING CMOS LOGIC

CMOS logic gates are implemented using complementary PMOS (pull-up) and NMOS (pull-down) transistor networks. PMOS conducts for logic 0 and NMOS conducts for logic 1, producing correct output levels. This design ensures low power consumption, high noise immunity, and efficient digital circuit operation.

4.2.1 CMOS Inverter

Digital electronics and integrated circuits implement the basic element of CMOS (Complementary Metal-Oxide-Semiconductor) inverters. From Figure 4.2, the structure incorporates arranged PMOS (P-type Metal-Oxide-Semiconductor) together with NMOS transistors to operate as a CMOS inverter.

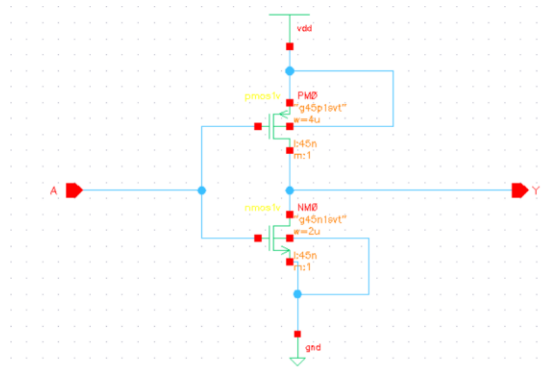


Fig. 4.2 Schematic Diagram of CMOS Inverter

4.2.2 CMOS NAND Gate

From Figure 4.3, the CMOS NAND gate contains two PMOS transistors connected in parallel and two NMOS transistors connected in series. The PMOS transistors are placed above, forming the pull-up network, while the NMOS transistors are placed below, forming the pull-down network.

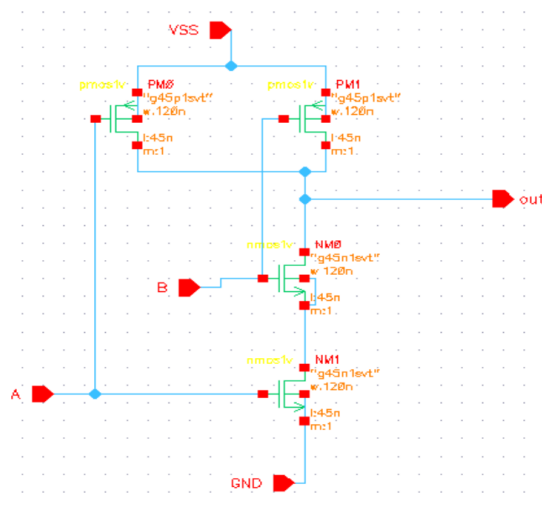


Fig. 4.3 Schematic Diagram of CMOS NAND Gate

The output becomes low when both inputs are high because the NMOS path conducts and pulls the output node to ground, while the PMOS transistors remain OFF. The output remains high when at least one input is low, since at least one PMOS transistor conducts and pulls the output to V_{DD} . Thus, the circuit produces a low output only when both inputs are high, which corresponds to the NAND logic operation.

4.2.3 CMOS NOR Gate

Figure 4.4, the CMOS NOR gate consists of two PMOS transistors connected in series and two NMOS transistors connected in parallel. The PMOS transistors form the pull-up network at the top, while the NMOS transistors form the pull-down network at the bottom. When both inputs are low, the PMOS transistors turn ON and the NMOS transistors remain OFF, resulting in a high output voltage. When either of the inputs is high, at least one NMOS transistor conducts and pulls the output to ground, while the corresponding PMOS transistor turns OFF. Thus, the output is high only when both inputs are low, which satisfies the NOR logic operation.

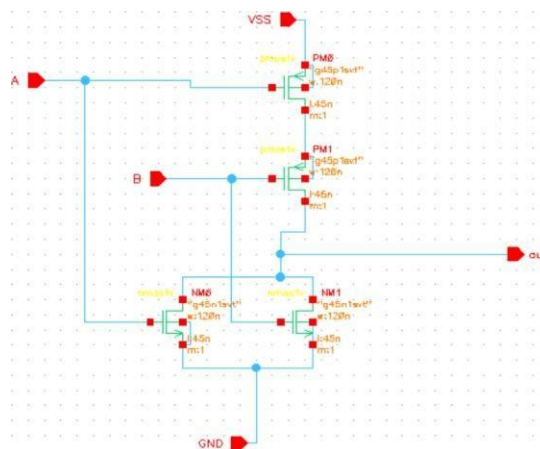


Fig. 4.4 Schematic Diagram of CMOS NOR Gate

4.3 LIMITATIONS OF CMOS LOGIC GATES

Speed: CMOS gates require more time to switch signals due to higher propagation delay compared to TTL logic gates.

Packing Density: Although CMOS achieves comparable or higher packing density than NMOS using pass-transistor logic, early CMOS implementations required more transistors, resulting in lower density.

Static Electricity Sensitivity: CMOS devices are sensitive to static charges, which can damage the circuits. Hence, protection mechanisms or proper handling techniques are required.

Cost: CMOS fabrication involves more processing steps compared to other logic

lies, leading to increased manufacturing cost.

Noise Susceptibility: Although CMOS has good noise immunity, it is still susceptible to noise, which may degrade signal quality.

Temperature Sensitivity: Variations in temperature can affect CMOS circuit performance, causing fluctuations in operation.

4.4 GDI TECHNIQUE

The Gate Diffusion Input (GDI) technique is an advanced digital design methodology used to implement low-power and area-efficient logic circuits. It is particularly suitable for modern VLSI systems where power consumption and silicon area are critical design constraints. As shown in Figure 4.5, the basic GDI cell resembles a conventional CMOS inverter structure but differs in its input configuration. It consists of three inputs: G (common gate input of PMOS and NMOS), P (input to the source of PMOS), and N (input to the source of NMOS). Unlike standard CMOS logic, where source terminals are connected to fixed supply voltages (V_{DD} and GND), the GDI technique allows these terminals to receive variable input signals.

This flexibility enables the implementation of multiple logic functions using a significantly reduced number of transistors. By applying different combinations of inputs to the G, P, and N terminals, various Boolean functions such as AND, OR, and multiplexers can be efficiently realized. For example, an AND gate can be implemented using only two transistors, whereas conventional CMOS requires more transistors for the same function. This reduction in transistor count leads to lower switching activity, reduced power consumption, and smaller chip area. Additionally, the GDI technique offers high-speed performance due to shorter propagation paths and reduced capacitance. However, the GDI technique also has certain limitations. It typically requires twin-well or silicon-on-insulator (SOI) technology for proper operation, which increases fabrication complexity. Moreover, threshold voltage drop and reduced output voltage swing may affect signal integrity. In this work, the GDI technique is used to design compact and power-efficient decoder circuits, which are integrated into SRAM arrays to improve overall performance.

4.4.1 Various Functions of GDI Cell

$$\text{OUT} = G'P + GN \quad (4.1)$$

From Equation 4.1, the GDI technique uses a basic cell with three inputs—G (gate), P (PMOS input), and N (NMOS input)—and produces an output. The GDI cell behaves

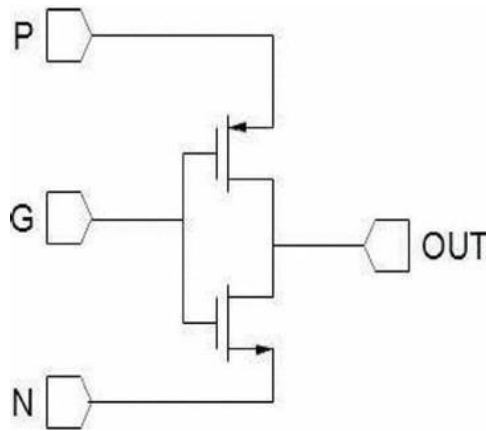


Fig. 4.5 Basic GDI Cell

like a multiplexer, where the output depends on the value of G . By assigning different input combinations to G , P , and N , various logic functions can be implemented efficiently using fewer transistors.

As shown in Table 4.1, the same GDI cell can realize inverter, AND, OR, XOR, XNOR, and MUX functions. This flexibility reduces circuit complexity, power consumption, and area, making GDI highly effective for low-power VLSI design.

Table 4.1 Functional Implementation of Logic Gates Using GDI Technique

G	P	N	OUT	FUNCTION
A	1	0	A'	INVERTER
A	B	0	$A'B$	Function1
A	1	B	$A' + B$	Function2
A	B	1	$A + B$	OR
A	0	B	AB	AND
A	B	C	$A'B + AC$	MUX
A	B	B'	$AB' + A'B$	XOR
A	B'	B	$AB + A'B'$	XNOR

4.5 FABRICATION CHALLENGES OF GDI TECHNIQUE

Although the Gate Diffusion Input (GDI) technique offers significant advantages in terms of reduced power consumption, transistor count, and area, its practical implementation faces several fabrication-related challenges. In conventional CMOS fabrication, the bulk (body) terminals of PMOS and NMOS transistors are connected to fixed potentials (VDD for PMOS and GND for NMOS). However, in GDI circuits, the source terminals receive variable inputs, which necessitates independent control of the bulk terminals to

avoid unwanted body bias effects. This requirement increases fabrication complexity and cost. Another major issue is the body effect, which arises due to the varying potential difference between the source and substrate. This leads to an increase in threshold voltage, affecting the switching characteristics of the transistors and degrading circuit performance. As a result, maintaining consistent operation across different input conditions becomes challenging. The GDI technique also suffers from reduced output voltage swing. Due to threshold voltage drops, the output may not reach full logic levels (i.e., exact VDD or GND), which can cause reliability issues, especially when cascading multiple logic stages. This limitation may require additional buffering circuits, partially offsetting the benefits of reduced transistor count. Furthermore, process compatibility is a concern, as standard CMOS fabrication processes are not always optimized for GDI-based designs. This can lead to difficulties in large-scale integration and limit its widespread adoption in industry. Another challenge is increased design complexity at the layout level. Since the inputs are applied to source terminals instead of fixed supply lines, careful layout planning is required to minimize parasitic effects and ensure proper isolation between wells.

4.6 MODIFIED GATE DIFFUSION INPUT (MGDI) TECHNIQUE

To address the fabrication and performance limitations of the conventional Gate Diffusion Input (GDI) technique, the Modified Gate Diffusion Input (MGDI) technique has been introduced as an improved design approach. MGDI retains the advantages of GDI while enhancing compatibility with standard CMOS fabrication processes. The primary improvement in MGDI lies in the modification of the input configuration. Unlike the original GDI cell, where inputs are applied to the source terminals of both NMOS and PMOS transistors, the MGDI technique applies inputs in a manner that ensures proper bulk-to-source connections. This eliminates the need for complex twin-well or triple-well fabrication requirements, making MGDI more suitable for standard CMOS technology.

As shown in Figure 4.6, MGDI helps in achieving full voltage swing at the output, overcoming one of the key drawbacks of the conventional GDI technique. One of the major advantages of MGDI is the reduction of the body effect. By maintaining appropriate biasing between the source and substrate, threshold voltage variations are minimized, leading to more stable transistor operation. This improvement eliminates the need for additional restoration circuits or buffers, thereby preserving the benefits of reduced transistor count and low power consumption.

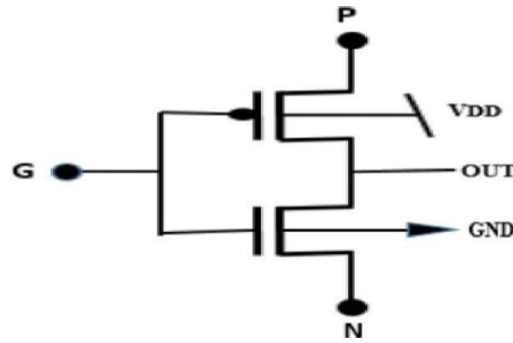


Fig. 4.6 Basic MGDI Cell

MGDI-based circuits into large-scale VLSI systems such as SRAM arrays and decoder circuits.

4.7 COMPACT DECODER

A GDI technique enables us to construct the decoder system which activates specific word lines and bit lines based on the provided address using decoders. The selected word line and bit line activate when receiving the provided address through 1x2 Decoder. This device includes one inverter and 2 AND gates. Decoding operations demand specific moments for corresponding selection of particular output paths. The core memory of A decoder functions as an addressing method for computer systems which uses hardware blocks for operation.

4.7.1 Implementation of AND Gate Using MGDI Technique

The memory address location contains single data entry points. The decreased transistor counts of AND gate designs minimize the total transistor count in the decoder, leading to fewer components. The reduction scheme transforms the original truth table into a simplified circuit. From Table 4.2 we can get Table 4.3 the reduced AND gate follows its truth table structure to design an optimized logic circuit.

Table 4.2 Truth Table of AND Gate

A	B	$Y = A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

The AND gate truth table can be simplified using don't-care conditions. When $A = 0$, the output is always 0 regardless of B . When $A = 1$, the output follows B .

Table 4.3 Reduction Table of AND Gate

A	B	$Y = A \cdot B$
0	x	0
1	x	B

This simplification helps in designing an efficient GDI-based AND gate. In operation, for $A = 0$, PMOS is ON and pulls the output to ground, giving 0. For $A = 1$, NMOS is ON and the output depends on input B .

Thus, Figure 4.7 the circuit produces the AND function with reduced complexity and transistor count.

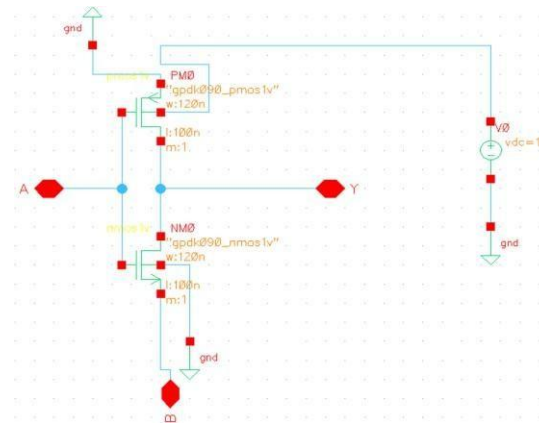


Fig. 4.7 Schematic Diagram of Novel AND Gate

4.7.2 CMOS vs MGDI Transistor Requirement Analysis

Table 4.4 presents a comparison of transistor count for various logic gates implemented using conventional CMOS and GDI techniques. It clearly shows that GDI-based designs require significantly fewer transistors than CMOS implementations. This reduction in transistor count leads to lower power consumption, smaller chip area, and improved speed. Hence, GDI is a more efficient approach for implementing logic gates in low-power VLSI design.

4.7.3 Implementation of Compact Decoder

Decoder: The term decoder refers to a combinational logic circuit in digital electronics that converts N -bit binary inputs into M output lines by activating only one output for each valid input combination. A decoder can produce a maximum of 2^N output lines and is widely used for address decoding and code detection.

Compact Decoder: A compact decoder is an optimized version of the standard decoder.

Figure 4.8 It performs the same decoding function—activating one of 2^N outputs for N

Table 4.4 Transistor Count Comparison of Logic Gates using CMOS and GDI

Gate	No. of Transistors (CMOS)	No. of Transistors (GDI)
AND	6	2
OR	6	2
NAND	4	2
NOR	4	2
XOR	12	4
XNOR	12	4

inputs—but with reduced hardware resources such as fewer transistors and logic gates.

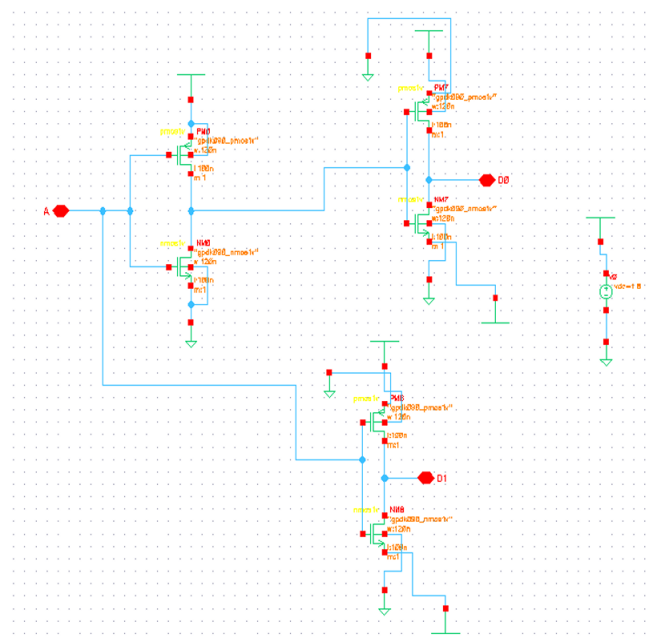


Fig. 4.8 Schematic Diagram of Compact Decoder

Compact decoders use techniques such as logic minimization and efficient gate arrangements to reduce the number of gates and interconnections. This results in lower power consumption and faster operation due to reduced logic levels. Additionally, compact decoders require less physical space, making them suitable for integration in large-scale VLSI systems such as SRAM arrays. Although compact decoders are slightly more complex to design, they are preferred in advanced applications where power and area efficiency are critical. In contrast, standard decoders are simpler to design but consume more power and area, making them less suitable for high-performance and low-power applications. As a result, they are less efficient and not ideal for modern high-performance and low-power VLSI applications compared to compact decoder designs.

4.7.4 Comparison of Compact Decoder and Standard Decoder

Decoders are essential components in digital systems, particularly in memory architectures such as SRAM. From Table 4.5 gives the comparison of Conventional decoders are typically implemented using standard CMOS logic, whereas compact decoders are designed using techniques like GDI/MGDI to achieve improved efficiency. The following comparison highlights the key differences between these two approaches.

Table 4.5 Comparison of Standard Decoder and Compact Decoder

Aspect	Standard Decoder (CMOS)	Compact Decoder (GDI/MGDI)
Transistor Count	High	Low
Power Consumption	High	Low
Area	Large	Compact
Speed (Delay)	Moderate	Faster
Voltage Swing	Full	Slightly reduced (improved in MGDI)

Key Improvements in MGDI:

- Better handling of threshold voltage degradation issues
- Reduced body effect problems
- Improved fabrication compatibility with standard CMOS processes
- Enhanced robustness under different process variations

4.8 ADVANCED APPLICATIONS

Arithmetic Circuits

- **Full Adder Designs:** Single-bit full adders can be implemented with as few as 10 transistors (compared to 28 in CMOS). Carry look-ahead techniques reduce the number of components, while ripple carry adders show improved power-delay performance.
- **Multiplier Architectures:** Array multipliers using GDI cells and Booth multipliers implement partial product generation efficiently. Wallace tree structures can also be realized using GDI/MGDI techniques for high-speed multiplication.

Memory Design

- **SRAM Cells:** GDI-based 6T and 8T SRAM cells improve power efficiency. Techniques exist to enhance read and write operations, and dual-port memory implementations are also possible.

Signal Processing Blocks

- **Multiplexers/Demultiplexers:** 2 : 1, 4 : 1, and 8 : 1 MUX designs can be implemented with significantly reduced transistor count, enabling high-performance selector circuits.
- **Comparators:** Magnitude comparators using GDI XOR implementations allow efficient multi-bit comparison with reduced complexity.

Recent Research Directions

- **Hybrid GDI-CMOS Design:** Combining GDI with CMOS technology provides improved performance. Strategic placement of GDI cells enhances efficiency.
- **FinFET Implementation:** GDI techniques can be adapted for FinFET technology. However, challenges such as body biasing limitations must be addressed to ensure proper functionality.

4.9 SUMMARY

Chapter 4 focuses on the design and implementation of an efficient compact decoder using advanced logic techniques to overcome the limitations of conventional CMOS-based designs. The chapter begins with an introduction to CMOS logic gates, explaining how NMOS and PMOS transistors form pull-up and pull-down networks to implement basic gates like inverter, NAND, and NOR. While CMOS offers low static power consumption, it suffers from drawbacks such as higher delay, larger transistor count, and increased area, which limit its efficiency in modern VLSI systems. To address these issues, the Gate Diffusion Input (GDI) technique is introduced as a low-power design method. GDI allows multiple logic functions to be implemented using fewer transistors, thereby reducing power consumption, delay, and chip area. However, it has challenges such as fabrication complexity, threshold voltage drop, and reduced output swing.

To overcome these limitations, the Modified GDI (MGDI) technique is proposed. MGDI improves compatibility with standard CMOS fabrication, ensures full voltage swing, reduces body effect, and enhances circuit reliability while maintaining low power and reduced transistor count. Finally, the chapter presents the design of a compact decoder using MGDI, where logic gates (especially AND gates) are optimized to minimize transistor usage. The compact decoder is compared with conventional decoders and is shown to achieve better power efficiency, reduced area, and improved speed, making it suitable for SRAM array applications and modern low-power VLSI systems.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 POWER ANALYSIS OF 2×2 SRAM ARRAY WITH DE- CODER AND COMPACT DECODER

The power analysis of the 2×2 SRAM array was carried out using both the conventional decoder and the proposed compact decoder across different technology nodes (45 nm, 90 nm, and 180 nm). The results clearly demonstrate a significant reduction in power consumption when the compact decoder is employed.

From Table 5.1, at 45 nm technology, the conventional decoder consumes $197 \mu W$, whereas the compact decoder reduces the power to $144 \mu W$, achieving a notable improvement. Similarly, at 90 nm, the power consumption decreases from $200 \mu W$ (standard decoder) to $149 \mu W$ (compact decoder). In the case of 180 nm, the power is reduced from $450 \mu W$ to $401.9 \mu W$. This improvement can be attributed to:

- Reduced transistor count in the compact decoder
- Lower switching activity
- Efficient utilization of the GDI (Gate Diffusion Input) technique

Additionally, as the technology scales down from 180 nm to 45 nm, total power consumption decreases due to reduced capacitance and supply voltage. However, regardless of scaling, the compact decoder maintains superior performance over the conventional design. In conclusion, integrating the compact decoder in the SRAM array significantly

Table 5.1 Comparative Power Evaluation of Standard and Compact Decoder in 2×2 SRAM Array

Technology	Power with Decoder	Power with Compact Decoder	Power Reduction	% Reduction
45 nm	$197 \mu W$	$144 \mu W$	$53 \mu W$	26.90%
90 nm	$200 \mu W$	$149 \mu W$	$51 \mu W$	25.50%
180 nm	$450 \mu W$	$401.9 \mu W$	$48.1 \mu W$	10.69%

enhances power efficiency without compromising functionality, making it suitable for low-power VLSI applications. In addition to power reduction, the compact decoder also contributes to improved speed performance. Due to fewer logic levels and reduced capacitance, the propagation delay is minimized, leading to faster word line activation.

5.2 POWER ANALYSIS OF 4×4 SRAM ARRAY WITH DE- CODER AND COMPACT DECODER

The power consumption analysis of the 4×4 SRAM array was performed using both the conventional decoder and the proposed compact decoder across 45 nm, 90 nm, and 180 nm technologies. The obtained results clearly indicate that the compact decoder significantly reduces power consumption in all cases. From the Table 5.2, At 45 nm technology, the power consumption decreases from 413 μW (standard decoder) to 339 μW using the compact decoder. For 90 nm, the power is reduced from 589 μW to 465 μW . A substantial improvement is observed at 180 nm, where the power drops from 932 μW to 701.9 μW . The reduction in power is more pronounced as the technology node increases, with the highest savings observed at 180 nm. This is mainly due to the efficient design of the compact decoder, which minimizes switching activity and reduces the number of transistors compared to the conventional decoder. Furthermore, when compared to the 2×2 SRAM array, the 4×4 array consumes higher power due to increased cell count and larger decoding circuitry. However, the compact decoder effectively mitigates this increase, making it a suitable choice for larger memory architectures. In conclusion, the proposed compact decoder demonstrates superior power efficiency across all technology nodes and array sizes. Its integration into SRAM design enhances overall performance, making it highly beneficial for low-power and 39 high-density VLSI applications.

Table 5.2 Comparative Power Evaluation of Standard and Compact Decoder in 4×4 SRAM Array

Technology	Power with Decoder	Power with Compact Decoder	Power Reduction	% Reduction
45 nm	413 μW	339 μW	74 μW	17.92%
90 nm	589 μW	465 μW	124 μW	21.05%
180 nm	932 μW	701.9 μW	230.1 μW	24.69%

5.3 COMPARATIVE TRANSISTOR COUNT EVALUATION OF CMOS AND GDI-BASED DECODERS

The transistor count comparison between conventional CMOS-based decoders and GDI-based compact decoders for 2×2 and 4×4 SRAM arrays is presented in Table 5.3. For the 2×2 SRAM array, the total number of transistors decreases from 86 (CMOS) to 68 (GDI), resulting in a reduction of approximately 20.93%. Similarly, for the 4×4

Table 5.3 Number of Transistors Used in 2×2 and 4×4 SRAM Arrays with Regular and Compact Decoder

SRAM Array	Decoder	Transistors (CMOS)	Transistors (GDI)	Reduction in Transistors	% Reduction
2×2 Array	1×2	86	68	18	20.93%
4×4 Array	2×4	220	188	32	24.24%

SRAM array, the transistor count is reduced from 220 (CMOS) to 188 (GDI), achieving a reduction of about 24.24%. This reduction is primarily due to the efficient implementation of logic functions using the Gate Diffusion Input (GDI) technique, which requires fewer transistors compared to traditional CMOS logic. As the SRAM array size increases, the advantage of using GDI becomes more prominent, leading to greater savings in hardware resources.

5.4 COMPARATIVE POWER ANALYSIS OF CMOS AND GDI- BASED SRAM ARRAYS

The power comparison between CMOS-based decoders and GDI-based compact decoders for different SRAM array sizes is presented in Table 5.4. The results clearly show that the GDI technique significantly reduces power consumption in both configurations.

From Table 5.4, for the 2×2 SRAM array, the power consumption decreases from 197 μW (CMOS) to 144 μW (GDI), achieving a reduction of approximately 25.05%. In case of the 4×4 SRAM array, the power is reduced from 413 μW to 339 μW , resulting in a 17.91% reduction. The higher percentage reduction in the smaller array indicates that GDI is highly efficient in compact designs. However, even for larger arrays, the GDI-based decoder maintains a considerable reduction in power, proving its scalability and effectiveness.

Table 5.4 Comparative Power Evaluation of CMOS and GDI-Based SRAM Arrays

SRAM Array	Decoder	Power (CMOS)	Power (GDI)	Power Reduction	% Reduction
2×2 Array	1×2	197 μW	144 μW	53 μW	25.05%
4×4 Array	2×4	413 μW	339 μW	74 μW	17.91%

This improvement is mainly due to:

- Reduced switching activity
- Lower transistor count
- Efficient logic implementation using GDI

Overall, the GDI-based decoder enhances power efficiency across different SRAM array sizes, making it suitable for low-power memory design in VLSI systems.

5.5 SUMMARY

Chapter 5 presents the simulation results and performance analysis of SRAM arrays using both conventional decoders and the proposed compact decoder. The evaluation is mainly based on key parameters such as power consumption, transistor count, and overall efficiency. The chapter begins with a power analysis of 2×2 and 4×4 SRAM arrays, comparing standard CMOS-based decoders with compact decoders. The results show that SRAM arrays using compact decoders consume significantly less power than those using conventional designs. This improvement is due to the reduced switching activity and lower transistor count in compact decoder circuits.

Next, a transistor count evaluation is performed between CMOS and GDI-based designs. The compact decoder requires fewer transistors, which directly contributes to reduced area and lower power consumption. This makes the design more efficient and suitable for large-scale integration. The chapter also includes a comparative power analysis of CMOS and GDI techniques, demonstrating that GDI-based implementations achieve better performance in terms of energy efficiency and speed. As technology scales down, the advantages of compact decoders become more prominent.

Overall, the results confirm that the proposed compact decoder outperforms the conventional decoder by providing lower power consumption, reduced area, and improved performance, making it highly suitable for modern low-power SRAM applications.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The analysis of SRAM arrays using the GDI-based compact decoder shows a clear improvement over conventional CMOS designs in terms of power and area efficiency. The 2×2 SRAM array achieves a significant power reduction of about 25.05%, indicating that the GDI technique is effective for both small and moderately larger memory structures. In addition to lowering power consumption, the proposed design also reduces the number of transistors, which helps in minimizing chip area and overall circuit complexity. Hence, the GDI-based decoder offers a balanced and efficient solution for designing low-power and area-optimized SRAM arrays in modern VLSI applications. Furthermore, the performance improvements achieved through the GDI technique highlight its suitability for scalable memory design.

As technology continues to advance toward smaller nodes and higher integration, reducing power and area becomes increasingly important. The consistent reduction in both parameters across different SRAM configurations demonstrates the robustness of the proposed approach. Therefore, the GDI-based compact decoder can be effectively utilized in future low-power VLSI systems.

6.2 FUTURE WORK

While this project has successfully demonstrated the benefits of compact decoders and GDI-based SRAM arrays, there is further scope for future work:

- **Scaling to Larger Arrays:** The current study focuses on 2×2 and 4×4 SRAM arrays. Future research can explore larger arrays (8×8 , 16×16 , 64×64), where optimization becomes more critical due to increased interconnect complexity and parasitic effects.
- **Layout-Level Optimization:** Physical layout design using Cadence Virtuoso Layout Editor can be performed to evaluate real-time area, routing complexity, and parasitic-induced delays, enabling accurate post-layout analysis.
- **FinFET-Based Implementation:** The proposed design can be extended to FinFET-based SRAM architectures to reduce leakage power and enhance performance, making it suitable for advanced VLSI applications.

Finally, the proposed architecture can be extended to real-time hardware implementation and tested for practical applications in embedded systems and cache memory.

- **Leakage Power Optimization:** Additional techniques can be applied to minimize leakage power, especially in deep submicron technologies.

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LIST OF PUBLICATIONS

PRESENTATIONS IN INTERNATIONAL CONFERENCES

- [1] Pratikhya Raut, S. Durga Bhavani, Y. Varshitha, Y. Poojitha, “Performance Analysis of Stability and Delay Trade-offs in 6T SRAM Cell at 45 nm and 90 nm” in *Proc. 1st IEEE International Conference on Recent Trends in Computing and Smart Mobility (RCSM)*, Bhopal, 2025.

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Performance Analysis of Stability and Delay Trade-offs in 6T SRAM Cell at 45 nm and 90 nm

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Abstract—The paper discusses the effect of CMOS technology scaling on the strength and speed of a standard 6T SRAM cell. Under the influence of varying supply voltages and ratios of transistor sizing, a comparative analysis is done on 45 nm and 90 nm nodes with Static Noise Margin (SNM) and write delay. Implementation and simulations Circuit Implementations are done in Cadence Virtuoso. The SNM is obtained in terms of DC properties through butterfly curves and the write delay is obtained through transient analysis. The results obtained indicate that the 90 nm cell is more stable with SNM of 287 mV at a voltage of 1.8 V, and 192.5 mV in the case of the 45 nm cell in this case. By comparison, the 45 nm design can operate faster with a write delay of 19.8 ps as VDD increases to 1.8 V; a comparison of VDD to 1.4 V shows the write delay increase of 39.8 ps, a duration half that of 45 nm design. The paper discusses the trade-off between performance and stability of nanoscale SRAM and provides valuable information to be used to choose technology nodes that are appropriate in the low-power and high speed memory system.

Index Terms— 6T SRAM Cell, Static Noise Margin (SNM), Access Delay, Technology Scaling, CMOS

I. INTRODUCTION

Static Random-Access Memory (SRAM) is a basic memory component of the current digital architectures, particularly in high-performance applications such as cache memories and in on chip tegra storage where access speed and low power consumption is extremely important. The most commonly used cell structure is the six-transistor (6T) SRAM cell with its compact architecture, power performance properties and long-term data storage reliability of bi nary data. The CMOS technology has been scaled progressively in terms of feature sizes of 180 nm down to 90 nm, and even 45 nm which have allowed a faster operation of the circuit, increased storage density, and reduced energy usage. In general, however, with small-scale device devices, small channel lengths have a negative effect on the device performance through the deterioration of both voltage stability and sensitivity to parameter variations. The effect of scaling on these two fundamental SRAM performance

metrics; Static Noise Margin (SNM) to determine reliability of stored information against unintentional noise, and access delay which determines the speed of the read/write process. Even though various papers are devoted to the advantages and drawbacks of the technology scaling to gain knowledge about the relationship between noise immunity and write performance in the advanced nodes is crucial in the design of the efficient memory systems design. Thus, the present work is aimed at a comparative analysis of the 6T SRAM behaviour at nodes 45 nm and 90 nm under the conditions of changes in cell ratio and the supply voltage at the input, and how the differences between these conditions impact SNM and write delay. This analysis offers useful rules in determining a suitable technology node that can be used to implement high-performance but low-power memory applications in nanoscale electronics.

II. LITERATURE REVIEW

Stability and performance of traditional 6T SRAM designs Scale-dependence of technology. Gadag and Chougla [1] investigated the effects induced by the size of the device on the power consumption of a typical CMOS-based 6T cell and revealed that the appropriate dimensioning of transistors may result in a substantial decrease of the total power consumption. Rath and Panda [2] also measured the behavior of SRAM under the various nodes in the technology and accentuated that scaling contributes to increasing the speed of memory processing at the expense of robustness. Rushi and Dasgupta [3] came up with an analytical frame of estimating Write Static Noise Margin (WSNM) in state-of-the-art nodes including 45nm and 65nm to have a better understanding of stability constraint during such write operations. Their results indicate that there is a growing threat of extrapolating data flipping as the transistors get smaller. This was supported by Saun and Kumar [4] who conducted comparative simulations that showed lower stability at lower CMOS process nodes because

of leakage and lower drive strength. Reddy et al. [5] provided efforts to improve area and reliability properties of SRAM, and they proposed a 6T implementation, which is compact and improves both the read and write stability and reduces silicon footprint. A comparison of SRAM designs at 180nm and 90nm was also conducted by Mutum and Mondal [6] and a constant trade-off was also found that stability reduces as the cell is scaled, and access time increases. In addition to more traditional CMOS, SRAM solutions of emerging device technologies have also been explored. Garidepalli et al. [7] discovered a low-power 4kb SRAM array based on 16nm GNRFTs to minimize dynamic power with the high performance. To augment this Mitta [8] examined CMOS variations of 6T SRAM at several technology nodes and found the need to control the relationship between transistor drive strengths to maintain read/write margins. Meshram et al. [9] suggested a high-speed SRAM platform that solicits low-power IoT platform, with the focus being on the efficiency of operation in miniature electronics. Recent works took a new direction to an interest in memory array architecture and variability issues. Namgiri [10] said that an 8×8 memory array on a 45nm node was analyzed and he checked that performance was improved with effective circuit-level optimizations. Even though the sources [11]–[13] are not focused on the design of SRAMs in particular, they provide some insights related to semiconductor-level models and details of device optimization that can significantly impact memory performance in scaled technologies. They consist of the enhancement of threshold voltage-dependent behavior and the reliability of conduction, applicable in the subsequent nanoscale SRAM designs. In general, it is well established in the literature that with the scaling of CMOS technology, a speedversusstability trade-off is to be faced in SRAM cells. Although delay performance is enhanced at smaller nodes, it seems to become more difficult to ensure the noise margins are sufficiently large enough - new sizing schemes and design level improvements are required.

III. SCHEMATIC OF 6T SRAM CELL

The 6T SRAM cell is a type of cell that has been popular in the memory array making it slim and with the colossal storage capacity of the information it bears. It operates under two cross coupled CMOS inverters that form a bistable latch used to form the binary information stored in the same manner. These NMOS access transistors require the connection of the storage nodes to the opposite (word-line (WL)) driven bit-lines (BL and BLB) ones. The assigning of the hold mode is given by the state of not reporting the access devices and retaining the logic of regenerative feedback of the inverters. As soon as a read or a write is invoked, the WL will make access path is available so that it can allow the updating or sensing of the information to take place. Transistor size is noteworthy so that it is the identical recording of the reads and write with the least use of power. The Cadence Virtuoso technology as 45nm technology node and 90nm technology node is the SNM and transient analysis of SNL 6T SRAM cell.

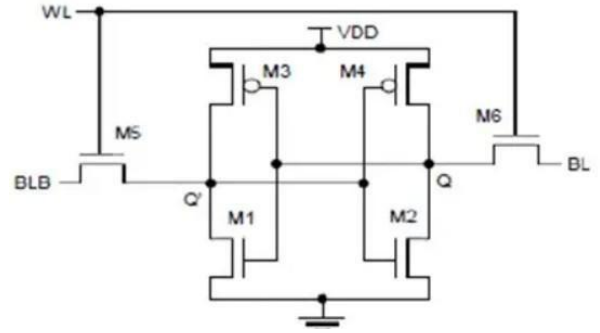


Fig. 1. Functional view of a typical 6T SRAM cell

IV. OPERATION OF 6T SRAM CELL

There are three fundamental conditions of operation of 6T SRAM cell which involve hold, read and write. These functions are based on the signal of the word-line (WL) and voltages across the bit-lines which are reversed to the bit-lines. The inverters are cross-coupled whereby, transistors that can be controlled are available to the memory array.

A. Hold Mode

During the hold mode, the WL is low and consequently, decouples the internal nodes to the bit-lines. The inverters as well have a positive feedback loop that enables storage of inverting logic state without dynamic refresh. It merely passes leakage currents and stability first of all, it is size dependent and secondly, it is dependent upon the supply voltage.

B. Read Mode

Pre Reading Before the real reading, the bit-lines are precharged to high logic level as in the case of the drive mode. When WL activation neurons develop under connecting the inner and bit lines, this occurs. The voltage across this bit-line at the 0 node across which the sense amplifier identifies the voltage in the face of a very small voltage. The access transistors must be overpowered by the power of the pull-down transistors in order to stop the read disturbances.

C. Write Mode

When a write is considered then the new data is dumped to BL and BLB on WL high. The bit-lines are supposed to be so strong that they may overturn the current state stored to have one of the nodes shifted into being low, and the other switching onto being high. A huge architecture thought processes then comes into the extension of transistor scaling eagerly in the process of writing of reality data.

V. DC ANALYSIS OF 6T SRAM CELL AND SNM EXTRACTION

DC analysis is widely employed to assess the stability of a 6T SRAM cell by quantifying its ability to retain data without corruption. The Static Noise Margin (SNM) is used as a performance metric, representing the maximum noise voltage tolerated before the stored bit flips. A high SNM indicates superior robustness against disturbances.

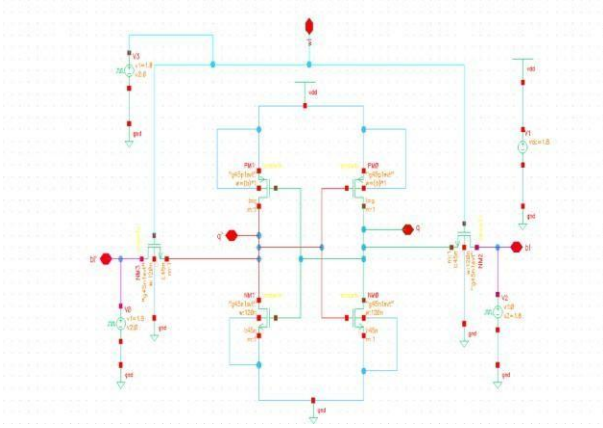


Fig. 2. Cadence Virtuoso schematic implementation of the 6T SRAM cell.

A. Butterfly Curve Method

To compute SNM, the voltage transfer characteristics (VTCs) of SNM results. Here, orchard plot of the voltage transfer characteristics (VTCs of cross-coupled inverters is obtained. The two inverter VTCs overlap with each other and form the butterfly shape of a graph. SNM can be defined as a distance of maximum square between the two lobes on the butterfly plot that are marked. The strategy allows making the comparison of other technology nodes, the supply voltages (V_{DD}), and the cell ratio (CR).

B. Cell Ratio (CR)

The transistor sizing ratio significantly impacts read stability. It is defined as:

$$CR = \frac{W_{PD}}{W_{Access}} \quad (1)$$

From equation [1] where W_{PD} is the width of the pull-down NMOS, and W_{Access} represents the width of the access NMOS transistor. A higher CR improves read stability by enhancing the pull-down strength, while a lower CR increases the likelihood of read disturbance.

C. Supply Voltage Effect

SNM is strongly influenced by V_{DD} . A higher supply voltage increases the drive strength of the cell transistors and enhances data stability. In contrast, reducing V_{DD} decreases the noise immunity, which is a concern for low-power SRAM applications. Therefore, DC analysis is performed under different supply voltages to ensure reliable operation across varying power levels.

VI. TRANSIENT RESPONSE AND WRITE DELAY

Transient analysis is used to examine the dynamic switching behavior of the 6T SRAM cell during write operations. When a write command is initiated, the bit-lines (BL and BLB) are driven to the appropriate logic levels, and the word-line (WL) is enabled to allow access through the NMOS transistors. The bit-line drivers must overcome the regenerative feedback of

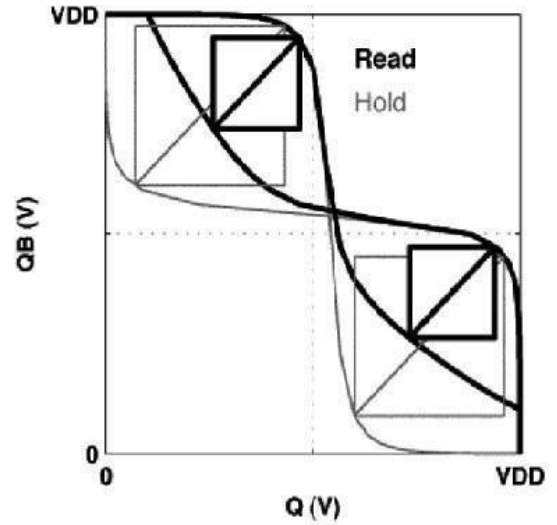


Fig. 3. Butterfly curves of the 6T SRAM cell during hold and read operations.

the cross-coupled inverters to successfully update the stored bit. Write delay is defined as the time difference between the activation of WL and the point at which the internal storage node reaches 50% of its target voltage state. This delay indicates the speed at which new data can be written into the cell. A higher supply voltage (V_{DD}) enhances the transistor drive strength, leading to faster node transitions and lower write latency. In contrast, when V_{DD} is reduced, the weakened drive capability results in slower switching and increased write delay. The technology node also influences write performance. Smaller geometries such as 45 nm provide reduced parasitic effects and shorter channel lengths, improving write speed compared to larger nodes like 90 nm. Hence, while scaling introduces stability challenges, it generally offers advantages in switching delay performance.

VII. RESULTS AND DISCUSSION

A. Effect of Supply Voltage on SNM during Hold Operation

Fig. 4 presents the SNM evaluation performed under the hold condition for the 6T SRAM cell. In this mode, the voltage transfer characteristics (VTCs) of the two cross-coupled inverters are plotted, and one curve is mirrored so that both overlap to form the characteristic butterfly pattern. The Static Noise Margin (SNM) is determined as the side length of the maximum square that can be fitted between the two lobes of the butterfly plot, representing the highest tolerable noise without altering the stored bit when the WL is inactive and the bit-lines remain isolated. Table I summarizes the variation of SNM with supply voltage (V_{DD}) for both 45 nm and 90 nm CMOS technologies. A decreasing trend is observed with voltage scaling. Specifically, the SNM of the 45 nm design reduces from 192.5 mV at 1.8 V to 96.6 mV at 1.4 V, whereas the 90 nm counterpart decreases from 287 mV to 236 mV across the same voltage range. Lower supply levels weaken transistor drive strengths in the pull-up and pull-

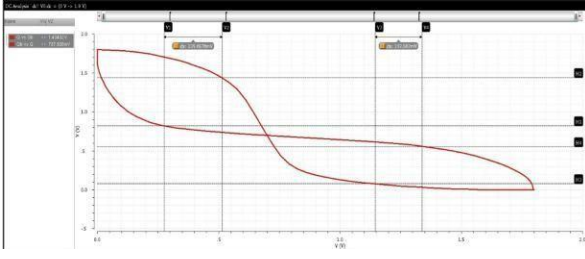


Fig. 4. Variation of Static Noise Margin (SNM) with supply voltage (V_{DD}) during hold operation for 6T SRAM cells obtained through DC analysis using butterfly curves.

down networks, making the stored data more vulnerable to electrical noise and fluctuations. It is also evident that the 90 nm SRAM cell consistently exhibits higher SNM values under all tested supply conditions. This improvement results from larger device geometries, reduced short-channel effects, and stronger voltage margins, which collectively enhance the robustness and noise immunity of the memory cell compared to the scaled 45 nm implementation.

TABLE I
VARIATION OF NOISE TOLERANCE WITH V_{DD} FOR NANOSCALE 6T SRAM CELLS FABRICATED IN 45 NM AND 90 NM PROCESSES.

V_{DD} (V)	SNM at 45 nm (mV)	SNM at 90 nm (mV)
1.8	192.5.0	287.0
1.6	140.0	254.0
1.4	96.6	236.0

B. Effect of Cell Ratio on SRAM Stability during Read Operation

Fig. 5 illustrates the SNM evaluation performed during the read operation of the 6T SRAM cell. When the word-line (WL) is enabled, the access NMOS transistors connect the internal nodes to the complementary bit-lines (BL and BLB). Under this condition, the stability of the stored data becomes dependent on the relative sizing of the pull-down and access transistors. As defined in Equation 1, the Cell Ratio (CR) determines the read capability. A larger CR strengthens the pull-down transistor, preventing unintended node voltage shifts caused by bit-line coupling. Conversely, a smaller CR can result in degraded read SNM and a higher risk of read disturbance as the access transistor begins to overpower the storage node. Table II presents the effect of CR variation on read SNM for both 45 nm and 90 nm implementations. The results show that increasing CR enhances read noise immunity in both technology nodes by improving the robustness of the pull-down path during bit-line interactions. Additionally, the 90 nm cell demonstrates consistently higher SNM values, which can be attributed to its larger transistors, reduced short-channel degradations, and stronger inverter drive capability.

C. Analysis of Write Delay Variation with V_{DD}

Fig. 6 is a transient simulation that can be used to get the write delay of the 6T SRAM cell. In asserting the word-

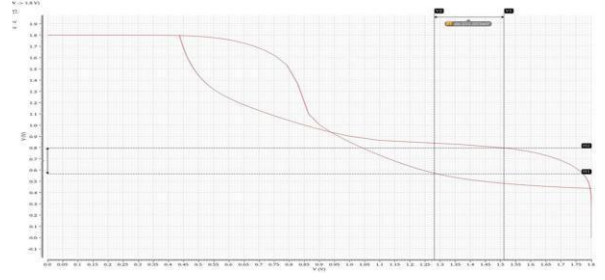


Fig. 5. Impact of cell ratio variation on read-state SNM for 6T SRAM structures implemented at 45 nm and 90 nm technologies.

TABLE II
SNM VARIATION WITH CELL RATIO (CR) DURING READ OPERATION FOR 6T SRAM CELLS AT 45 NM AND 90 NM NODES.

Cell Ratio (CR)	SNM at 45 nm (mV)	SNM at 90 nm (mV)
1.8	231.13	295.69
1.6	202.14	229.0
1.4	198.0	213.1

line (WL), the access transistors make connection between the internal storage nodes and the bit-lines (BL and BLB), and this allows the write operation. The delay time between the actual activation of WL and the time when the storage node voltage has attained 50 percent of its final value. Table III shows the write delay under various supply voltages (V_{DD}) of the 45nm and 90nm technology node. The higher V_{DD} , the lower the delay of both nodes as a result of enhancement of transistor drive strength and switching speed. As an example, in the 45nm design the delay decreases between 19.8ps at 1.4V, and 12.18ps at 1.8V. Similarly, the cell with 90nm depicts a decrease in 46.8ps to 38.01ps across the voltage range. These findings demonstrate that the scaled 45nm technology has higher write performance than 90nm node.

TABLE III
WRITE DELAY VALUES FOR 6T SRAM CELLS IN 45NM AND 90NM TECHNOLOGIES UNDER DIFFERENT SUPPLY VOLTAGES.

V_{DD} (V)	45nm (ps)	90nm (ps)
1.8	31.12	38.01
1.6	34.3	42.6
1.4	39.8	46.8

VIII. PERFORMANCE ANALYSIS

Based on the comparison Table IV The performance achieved by the 6T SRAM cell was evaluated in terms of stability of hold, read strength, and write delay of the 45nm and 90nm CMOS process nodes. The effect of the supply voltage (V_{DD}) and cell ratio (CR) have been studied and learnt about the trade-offs in the design of nanoscale memories. Hold SNM and (V_{DD}) are related because the greater (V_{DD}) the further the better drive strength of the pull-up and pull-down transistors and guarantees noise immunity when retaining data. The cell ratio is mainly controlled by the read stability as a higher CR offers a greater pull-down capacity to avoid

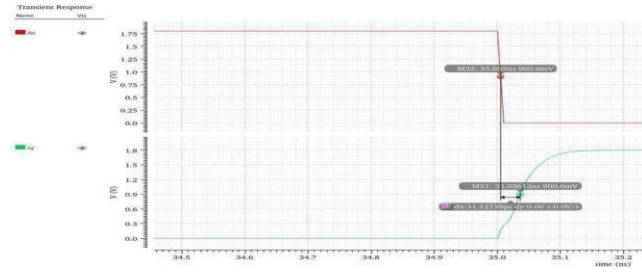


Fig. 6. Transient simulation waveform illustrating the write operation of the 6T SRAM cells.

TABLE IV
PERFORMANCE COMPARISON SUMMARY OF 6T SRAM CELLS AT 45 NM
AND 90 NM TECHNOLOGY NODES.

Metri	Variation	Observation	Impact / Remarks
Hold SNM	V_{DD} variation	Higher V_{DD} improves SNM, lower V_{DD} reduces SNM	Affects data retention and cell stability during hold operation
Read SNM	Cell Ratio (CR) variation	Higher CR increases read SNM, lower CR reduces read SNM	Determines read stability; prevents read disturb at optimal CR
Write Delay	V_{DD} variation	Higher V_{DD} reduces write delay, lower V_{DD} increases delay	Influences write speed and reliability; trade-off with power

disruptions caused by the bit-lines during the read access. On the other hand, higher CR predisposes the chance of erroneous flipping of data. The lower the V_{DD} , the lower the write delay is because the larger switching capability minimizes the amount of time needed to update stored information. The 45nm technology node has high write speed because of reduced parasitic effects and smaller transistor size, and the 90nm node is always able to offer the high SNM and the stabilized node because of the reduced short-channel effects. All in all, the findings show that there is a trade-off of stability versus speed between the two nodes of technology. The 45nm SRAM cell is preferable in the high performance applications where it is preferred in applications where rapid write operations are important, the 90nm implementation is more preferable in noisy systems where the low data retention properties are required.

IX. FUTURE WORK

In the future, future employees will strive to advance performance studies on 6T SRAM cells by considering other data such as read delay, power consumption and area usage as a way of achieving a more precise analysis in the advanced technology node. Simulations will be done using Monte Carlo and corner analysis of PVT (Process-Voltage Temperature) will be done to determine the effect of process variation, altering temperature and voltage variability with reference to total reliability. In addition, the various SRAM architecture (8T and 10T cells) will have to be checked in order to assess the enhanced stability of the data and the strength of the read/write operations.

X. CONCLUSION

The trade off between stability and write performance of the 6T SRAM cells grow under 45nm and 90nm CMOS technology was explored by conducting a comparative study. The Static Noise Margin (SNM) and write delay were examined with the modification of supply voltage (V_{DD}) and ratio of cells (CR). The results indicate that the data retention capacity and switching speed is on the rise as (V_{DD}) is increased, but the immunity to noise and the write latency are lower and higher respectively, as (V_{DD}) decreases. The study also indicates that CR makes the largest contribution towards read stability since the pull-down transistors that occur will be more powerful, and helps in countering the read disturb condition. The scaling of technological performance has a mixed effect: the 45nm node can be made to work faster since the parasitic effects are reduced, and the 90nm node can provide superior stability to the voltage conditions. This is the reason, V_{DD} and CR should be chosen correctly and make sure that high-performance SRAM systems can work effectively and be stable.

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SDG and its Target Mapping

Project Title: Performance Analysis of Highly Efficient Compact Decoders for SRAM Arrays

S No	Relevant SDG	SDG Target	Mapping Level	Justification
1	SDG 9: Industry, Innovation and Infrastructure	Target 9.5 – Enhance scientific research and upgrade technological capabilities of industrial sectors, encouraging innovation	3	The project focuses on designing efficient SRAM arrays and compact decoders using advanced VLSI techniques. It promotes innovation in semiconductor memory design, contributing to technological advancements in embedded systems and digital electronics.
2	SDG 7: Affordable and Clean Energy	Target 7.3 – Improve energy efficiency	3	The proposed compact decoder design reduces power consumption in SRAM arrays. Low-power memory design directly contributes to energy-efficient electronic systems and sustainable computing solutions.
3	SDG 11: Sustainable Cities and Communities	Target 11.6 – Reduce environmental impact of cities	2	Energy-efficient SRAM circuits reduce overall power usage in electronic devices, indirectly minimizing energy demand and environmental impact in urban infrastructure systems.
4	SDG 12: Responsible Consumption and Production	Target 12.2 – Achieve sustainable management and efficient use of natural resources	2	By reducing area and power in SRAM design, the project supports efficient utilization of semiconductor resources, minimizing material usage and improving chip efficiency.

This project presents the design and performance analysis of compact decoders for SRAM arrays using efficient VLSI techniques. The main contribution lies in reducing power consumption, delay, and area while improving memory performance. Hence, it is strongly connected to **SDG 9, SDG 7, and SDG 11**, with additional indirect contributions to environmental sustainability and education.

PO- PSO mapping

PO mapping

Performance Analysis of Highly Efficient Compact Decoders for SRAM Arrays	
Understanding the fundamentals of SRAM cell operation, including hold, read, and write mechanisms using 6T architecture	1:2
Analyzed SNM, VTC curves, and butterfly curves to evaluate cell stability under different voltages and cell ratios.	2:2
Identified the impact of technology scaling (45 nm & 90 nm) on SRAM performance, delay, leakage, and stability.	3:2
Designed and simulated 6T SRAM cell, decoder, sense amplifier, and write driver using Cadence Virtuoso.	4:1
Used modern EDA tools such as Cadence Virtuoso and Spectra simulators, learning schematic design, simulation, and waveform analysis.	5:3
Observed how compact decoder design can reduce area and power, improving overall memory efficiency.	6:1
Recognized how optimized SRAM circuits can support low-power VLSI applications such as IoT and embedded systems.	7:1
Followed professional ethics and proper engineering practices during simulation and reporting.	8:2
Worked effectively as a team, coordinating tasks such as schematic design, simulation, analysis, documentation, and integration.	9:3
Understood multiple review processes, tested and validated simulation outcomes for SRAM array operation	10:2
Managed project timelines and resources effectively to complete all simulations and documentation.	11:1
Gained awareness of future research scope such as compact GDI decoders, FinFET SRAMs, and large memory arrays	12:2

PSO mapping

Performance Analysis of Highly Efficient Compact Decoders for SRAM Arrays	
Designed and analyzed a 6T SRAM cell and 2×2 SRAM array using simulation tools, demonstrating strong domain-specific circuit design skills.	1:3

Guide Signature

Project Review Panel

Performance Analysis of Highly Efficient Compact Decoders for SRAM Arrays

Students Details:

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1. S. Durga Bhavani	228W1A0457
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Batch No:	A02	AY 2022-2026
Guide Name	: Dr. Pratikhya Raut	
Designation	: Assistant Professor	

Abstract:

This project analyzes a compact decoder for SRAM arrays using the GDI technique to overcome the limitations of conventional CMOS decoders, such as high power, delay, and area. A 6T SRAM cell and 2×2 and 4×4 arrays are designed and simulated in Cadence Virtuoso. The proposed GDI-based decoder reduces transistor count, power consumption, and delay, and is evaluated across 180 nm, 90 nm, and 45 nm technologies. Results show improved power efficiency, speed, and area, making it suitable for low-power, high-performance VLSI applications.

Keywords: SRAM Array, 45 nm/90 nm/180 nm Technology, Compact Decoder, GDI Technique, Low Power VLSI, Transistor Count Reduction

Introduction:

SRAM is widely used in modern digital systems due to its high speed, low latency, and reliability. An SRAM array includes memory cells and peripheral circuits, where the decoder plays a key role in cell selection. However, conventional CMOS decoders suffer from high power, delay, and area, especially with technology scaling. This work proposes a GDI-based compact decoder to reduce transistor count, power consumption, and delay. A 6T SRAM cell and 2×2 and 4×4 arrays are designed using Cadence Virtuoso and analyzed across 180 nm, 90 nm, and 45 nm technologies. Results show improved power efficiency, speed, and area for low-power VLSI applications.

Objectives:

- Design and simulate a 6T SRAM cell and SRAM arrays
- Develop a GDI-based compact decoder for low-power operation.
- Compare CMOS and GDI designs in terms of power, and area.
- Analyse performance across 180 nm, 90 nm, and 45 nm technologies.

Methodology:

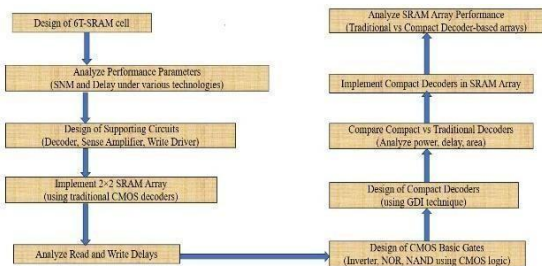


Fig. 1: Proposed Methodology

Proposed Architecture:

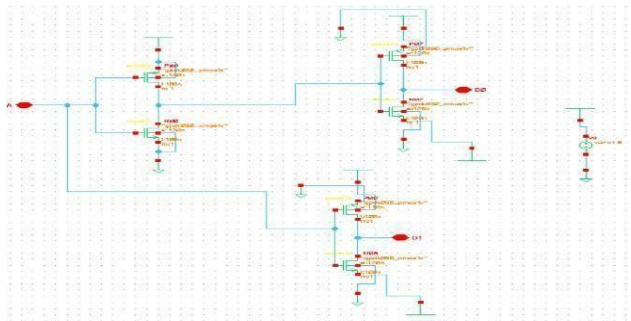


Fig 2: Compact Decoder

Results:

SRAM Array	Decoder	Power (CMOS)(μW)	Power (GDI) (μW)	Power Reduction (μW)	% Reduction
2×2 Array	1×2	197	144	53	25.05%
4×4 Array	2×4	413	339	74	17.91%

Table 1: Transistor Count Reduction of SRAM Arrays

SRAM Array	Decoder	Transistors (CMOS)	Transistors (GDI)	Reduction in Transistors	% Reduction
2×2 Array	1×2	86	68	18	20.93%
4×4 Array	2×4	220	188	32	24.24%

Table 2: Power Reduction of SRAM Arrays

The comparison shows that GDI-based decoders significantly reduce transistor count and power consumption compared to conventional CMOS designs. For the 2×2 SRAM array, transistor count is reduced by 20.93% and power by 25.05%, while the 4×4 array shows a reduction of 24.24% in transistors and 17.91% in power.

Conclusion:

This work presents the design and analysis of a GDI-based compact decoder for SRAM arrays to overcome the limitations of conventional CMOS designs. A 6T SRAM cell and 2×2 and 4×4 arrays were successfully implemented and evaluated using Cadence Virtuoso across 180 nm, 90 nm, and 45 nm technologies. The results demonstrate significant reductions in power consumption, transistor count, and delay, confirming the efficiency of the GDI technique. Overall, the proposed design provides a scalable and effective solution for low-power, high-performance SRAM architectures in modern VLSI systems.